RESEARCH

Björn Friedrich

Empowering Independent Living using the ICF

An Unobtrusive Home Monitoring Sensor System for Older Adults

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An Unobtrusive Home Monitoring Sensor System for Older Adults

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Oldenburg, Germany

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Abstract

This thesis introduces a system that continuously keeps track of the functional status of older adults through monitoring their behaviour, physical parameters, and mobility in their domestic environments in daily life. Functional decline in older adults can lead to a loss in independence and an increased need of care. As a consequence, moving to a nursing home may be indicated. Moving to a nursing home may has a negative impact on the three innate psychological needs of humans For the purpose of enabling older adults independently living in their own homes for longer, the system links data from unobtrusive and privacy preserving ambient and wearable sensors to five items of the *International Classification of Functioning, Disability and Health* (ICF), developed by the *World Health Organization* (WHO), from three categories and measures their change over time. The linking was realised by one Deep Neural Network (DNN), linear regression models, and a new unsupervised concept drift detection algorithm which combined a Variational Autoencoder (VAE) with a statistical hypothesis test. Based on the information provided by the system, health care professionals can design individualised rehabilitation programmes and monitor their effect. Moreover, the activities of daily living where assistance is needed can be identified and pointed assistance can be provided. Data from 20 (pre–)frail older adults (aged $\geq$75y) collected during a 10–month observational randomised pilot intervention study was used for evaluation. The DNN achieved an accuracy of 94.27 % and 95.79 % on predicting the *Short Physical Performance Battery* (SPPB) and *Timed Up & Go* (TUG) score respectively. The linear regression model was able to detect all significant weight changes related to malnutrition and all abnormal days were correctly recognised by the unsupervised concept drift detection algorithm and hence the system provides useful information for health care professionals.

Kurzzusammenfassung

In dieser Arbeit wurde ein System zur kontinuierlichen Beobachtung des funktionalen Zustands älterer Menschen im täglichen Leben entwickelt. Das System beobachtet das Verhalten, physische Parameter und die Mobilität im Bereich des täglichen Lebens. Funktionaler Abbau im Alter führt zu einer erhöhten Abhängigkeit bis hin zu einem notwendigen Umzug in eine Einrichtung des betreuten Wohnens oder ein Pflegeheim. Insbesondere letzteres kann negative Auswirkungen auf die drei psychologischen Grundbedürfnisse von Menschen haben. Das System kann in den häuslichen Umgebungen von älteren Menschen installiert werden, um es ihnen zu ermöglichen länger dort zu leben. Es verbindet Daten von Hausautomations-, Energie- und tragbaren Sensoren, die die Privatsphäre nur minimal beeinträchtigen, mit fünf verschiedenen Kategorien aus drei Kapiteln des *Internationale Klassifikation der Funktionsfähigkeit, Behinderung und Gesundheit* (ICF) der *Weltgesundheitsorganisation* (WHO). Die Kategorien und ihr Verlauf werden mit Hilfe eines tiefen neuronalen Netzes, einem Regressionsmodells und einem neu entwickelten Algorithmus, der ein probalistisches neurales Netz mit einem statistischen Hypothesentest kombiniert gemessen. Diese Informationen kann geschultes medizinisches Personal nutzen, um individualisierte Rehabilitationsmaßnahmen zu konzipieren, einzuleiten und ihren Effekt zu überprüfen. Außerdem kann das System Situationen, in denen ältere Menschen Unterstützung benötigen, identifizieren, damit gezielt unterstützt werden kann. Mit den Daten von 20 über 75–jährigen und gebrechlichen Teilnehmern*innen, mit Frailty Syndrom oder im Vorstadium (pre–frail), einer 10–monatigen randomisierten Interventionspilotstudie wurde das System evaluiert. Das tiefe neuronale Netz erreichte eine Genauigkeit von 94,27 % und 95,79 % bei der Vorhersage von *Short Physical Performance Battery* (SPPB) und *Timed Up & Go* (TUG) Ergebnissen und das Regressionsmodell war in der Lage alle signifikanten

auf Mangelernährung hindeutenden Gewichtsveränderungen zu erkennen. Außerdem konnte der neu entwickelte Algorithmus alle von der Norm abweichende Tage korrekt identifizieren. Die Ergebnisse haben gezeigt, dass es wertvolle Informationen für medizinisches Personal bereitstellt.

Contents

1 Introduction 1

2 A Deep Learning Approach for TUG and SPPB Score Prediction of (Pre–) Frail Older Adults on Real—Life IMU Data 5

3 Detecting Impending Malnutrition of (Pre–) Frail Older Adults in Domestic Smart Home Environments 31

4 Using Sensor Graphs for Monitoring the Effect on the Performance of the OTAGO Exercise Program in Older Adults 55

5 Unsupervised Statistical Concept Drift Detection for Behaviour Abnormality Detection 75

6 A System for Monitoring the Functional Status of Older Adults in Daily Life 97

7 General Discussion 119

Bibliography 129

List of Figures

Figure 2.1 The deep neural network used for this research. Each sensor modality had its own input. The inner structure of the blocks are shown in Figure 2.2 13
Figure 2.2 The blocks of the deep neural network. Details about the blocks and the layer parameters can be found in the Tables A1–A6 in Appendix A 14
Figure 2.3 The ROC curve of the TUG model 16
Figure 2.4 The ROC curve of the SPPB model 17
Figure 2.5 The loss of the TUG model. For the first 25 epochs the loss indicates that the learning rate is slightly too large. From epoch 25 the progress shows an asymptotic behaviour 17
Figure 2.6 The loss of the SPPB model. For the first 25 epochs the loss indicates that the learning rate is slightly too large. From epoch 25 the progress shows an asymptotic behaviour 18
Figure 2.7 The accuracy of the TUG model. According to the progress of the loss, the accuracy fluctuates in the first 25 epochs and shows an asymptotic behaviour after epoch 25. The best validation accuracy score was 95.89% at epoch 52 19
Figure 2.8 The accuracy of the SPPB model. According to the progress of the loss, the accuracy fluctuates in the first 25 epochs and shows an asymptotic behaviour after epoch 25. The best validation accuracy score was 94.29% at epoch 67 20

Figure 3.1 The flats of two participants of the OTAGO study. The layout and size of the flats are different and so are the installed sensors. In the right flat a great variety of sensors were installed, especially in the kitchen and the hallway 36
Figure 3.2 The progress of the body weight of the used subcohort over the whole study duration. Gaps in the graphs indicate missing values. The numbers in the legend are the participant IDs 36
Figure 3.3 The first month of participant 1. Gain in body weight (61.9–69.0 kg), decrease in HGS (24.0– 23.0 kg), increase in the KPI (≈10–15 min). HGS: Hand Grip Strength, KPI: Key Performance Indicator. 46
Figure 3.4 The sixth month of participant 1. Gain in body weight smaller than 1% (69.5–70.0 kg) and considered as no change, decrease in HGS (22.0–21.7 kg), nearly constant KPI (≈12–13 min). HGS: Hand Grip Strength, KPI: Key Performance Indicator. 47
Figure 3.5 The second month of participant 4. Significant loss in body weight (54.1–48.8 kg), decrease in HGS (12.0–11.3 kg), increase in the KPI (≈39–76 min). HGS: Hand Grip Strength, KPI: Key Performance Indicator. 47
Figure 3.6 The third month of participant 4. Loss in body weight (48.8–47.4 kg), decrease in HGS (11.3– 11.0 kg), increase in the KPI (≈36–88 min). HGS: Hand Grip Strength, KPI: Key Performance Indicator. 48
Figure 4.1 The flat of participant 6. The boxes indicate the position of the motion sensors, and the shades the recorded area 63
Figure 4.2 The sensor graph of the flat (Figure 4.1) of participant 6 with the baseline weights 64
Figure 4.3 The fitted lines of the relative SPPB scores in points. The scale on the *y*-axis is the MCID 67
Figure 4.4 The fitted lines of the relative TUG time in s 67
Figure 4.5 The fitted lines of the difference in % of the daily sensor graphs 68
Figure 5.1 A flat of a participant of the OTAGO study 80

Figure 5.2 One example of an activity probability map of a kitchen. Hours 5 and 6 show high probabilities for using the toaster, the hot plate and the kettle 81
Figure 5.3 One example of an activity probability map of a living room. The power consumption sensor was attached to a multi socket where a lamp and a radio were connected to 82
Figure 5.4 The illustration of our unsupervised statistical concept drift detection approach 83
Figure 5.5 The illustration of our VAE. The used activation function was the sigmoid function and the used optimiser was ADAM with an initial learning rate of 1×10^{-4} and an exponential decay after the 5th epoch. The training was automatically stopped, if no improvement of the validation loss larger than 0.001 was found within 5 epochs 85
Figure 5.6 The four typical drift types we used for validating our method 86
Figure 5.7 The drift of the distributions of participant 1. After beginning the chemotherapy no normal day is found using 1σ for testing the H_0. The left histograms show the distributions at the beginning and the right at the end of the study 88
Figure 5.8 The drift of the distributions of participant 16. After beginning the chemotherapy no normal day is found using 1σ for testing the H_0. The left histograms show the distributions at the beginning and the right at the end of the study 89
Figure 6.1 An illustration of one of the flats, which was equipped with the sensor system 101
Figure 6.2 The disease model used as foundation of the ICF[15] 102
Figure 6.3 The final system comprised of three components 103
Figure 6.4 Activity probability map of participant 1's kitchen for day 1 [34] 105
Figure 6.5 The system outputs of Mr. Mueller for month 0–9 107
Figure 6.6 The system outputs of Mrs. Winter for month 0–9 108

List of Tables

Table 2.1 The baseline characteristics of the study cohort 10
Table 2.2 The characteristics at the end of the study cohort 10
Table 2.3 This table shows the number of windows for each class of the SPPB and the TUG assessments. The range of the TUG score is smaller than the range of the SPPB score 12
Table 2.4 This table shows the number of windows for participant 16 who was excluded from the training set 12
Table 2.5 This table shows the number of windows for participant 19 who was excluded from the training set. Score 2 of the SPPB were not considered for evaluation, because the model for the SPPB was not trained with class 2 12
Table 2.6 The accuracy on the test set and the two excluded participants 20
Table 2.7 The specificity and sensitivity for each TUG score on the test set 20
Table 2.8 The specificity and sensitivity for each considered SPPB score on the test set 21
Table 2.9 The confusion matrix of the TUG model. The class with the least false classifications is 1 and the class with the most false classifications is 3. t = true label, p = predicted label 21

Table 2.10 The confusion matrix of the SPPB model. The class with the least false classifications is 3 and the class with the most false classifications is 9. t = true label, p = predicted label 21
Table 2.11 The confusion matrix of the TUG for participant 16. The participant scores 2 and 3, but not 1. The most values are in class 2. Due to the imbalance the accuracy is high, but the performance is low. t = true label, p = predicted label 22
Table 2.12 The confusion matrix for SPPB of participant 16. The only class with correct predictions is class 4. t = true label, p = predicted label 22
Table 2.13 The confusion matrix for TUG of participant 19. The model is not able to distinguish between score 3 and score 2 for this participant. t = true label, p = predicted label 22
Table 2.14 The confusion matrix for SPPB of participant 19. This participant was the only one with a SPPB score of 2. Since the model was not trained to classify this score, the values for score 2 were not considered for classification. t = true label, p = predicted label 23
Table 3.1 The characteristics of the study cohort at baseline (T0) 37
Table 3.2 The characteristics of the study cohort at the end (T10) ... 37
Table 3.3 The characteristics of the used subcohort at baseline (T0) 38
Table 3.4 The characteristics of the used subcohort at the end (T10) 38
Table 3.5 The results of the correlation analysis for participant 1 42
Table 3.6 The results of the correlation analysis for participant 2 42
Table 3.7 The results of the correlation analysis for participant 3 43
Table 3.8 The results of the correlation analysis for participant 4 43
Table 3.9 The results of the correlation analysis for participant 5 43
Table 3.10 The results of the correlation analysis for participant 6 44
Table 3.11 The results of the correlation analysis for participant 7 44
Table 3.12 The results of the correlation analysis for participant 8 45
Table 3.13 The results of the correlation analysis for participant 9. 45
Table 3.14 The results of the correlation analysis for participant 10 ... 46
Table 4.1 The baseline/end characteristics of the cohort 60

Table 4.2 The baseline/end characteristics of the used sub-cohort at the beginning of the study 60
Table 4.3 The matrix is the baseline adjacency matrix of participant 6 62
Table 4.4 The adjacency matrix of a random day of participant 6 63
Table 5.1 The baseline characteristics of the cohort 79
Table 5.2 The characteristics of the cohort at the end of the study . . . 79
Table 5.3 The results of the distribution approximation. The mean and variance rows show the mean and variance of the difference between the baseline {0,1} and the approximations 87
Table 5.4 The results of the concept drift validation 88
Table 5.5 The results of our statistical concept drift detection algorithm on the real–world dataset for different σ. normal days / abnormal days 90
Table 6.1 Characteristics of the study cohort at baseline (T0) (edited [31]) 100
Table 6.2 Characteristics of the study cohort at the end (T10) (edited [31]) 100
Table 6.3 The linking of the assessment scores to the ICF scores 104
Table 6.4 The available components, i.e. ICF items, for each participant 109



Introduction

1

Older adults are prone to functional decline and eventually need support and extensive care by relatives or professional caretakers like nurses. In cases where the family or nurses cannot provide sufficent support at home, older adults have to relocate to a nursing home or an assisted living scheme. Moving to a nursing home affects the three innate psychological needs—namely competence, autonomy, and relatedness [23, 6, 24]. Additionally, the cost of a nursing home is a financial burden on the older adult, the relatives, and the health care system itself. In the industrial nations, particularly in Germany and Japan, the number of older adults increases and so does the number of people in need of care [5, 17]. Accordingly, the pressure on the health care system will be rising in the near future. Moreover, the lack of professional caretakers and nurses affects the quality of care [4, 20]. Enabling older adults living longer independently in their own domestic environments helps them to maintain a good condition of the innate psychological needs and releases the financial burden on relatives and the health care system. Utilising low–cost unobtrusive and privacy preserving home automation, power consumption, and wearable sensors for continuously monitoring the functional status of older adults could be solution to identify the domains where assistance is needed early and initiate the appropriate interventions to prevent a loss in independence. Furthermore, the current level of care of care–dependent older adults could be identified and the care plan could be made more effective by tailoring in to the individual. Eventually, the system enables older adults living an independent life in their own homes instead of moving to a nursing home. In spite of affecting the privacy, these types of sensors are well accepted for the purpose of keeping independence among older adults in different cultures [7, 13, 14, 25]. The sensor data enables medical professionals to get deeper

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insights of where the older adult needs support, to design individualised rehabilitation programmes, and monitor the rehabilitation progress. This leads to better rehabilitation outcomes and lessens the impact of medical condition on the functional status. One important advantage of unobtrusive monitoring systems are the unbiased measurements in daily life. People tend to give their best effort in testing situations and the results rather reflect their capacity than their performance [1]. Especially in the case of older adults, it holds that capacity is not performance [10]. This fact is important because real– life measurements tend to be better predictors than assessment measurements—e.g. self–selected gait speed is a valuable predictor for mortality rates and daily ambulatory activity [26, 21, 15]. Therefore, a prototype of a system for monitoring the functional status of older adults in daily life is introduced. The system links sensor data to five items of three categories of the *International Classification of Functioning, Disability and Health* (ICF) [19]. The items and categories were taken from the *Core Set: Geriatrics Comprehensive* [3]. The system is extendable by design because older adults (i.e. geriatric patients) have a variety of medical conditions with complex interactions [18, 29]. A preliminary data analysis showed that this was also visible in the data [9]. The complexity of medical conditions and their interactions also requires individualised systems and, taking that into account, the proposed system was constructed using the data of each individual separately. The system was composed of three components, where components 1 and 3 were made up of two sub–components.

Component 1 measured the mobility based on Inertial Measurement Unit (IMU) data and a deep neural network. The input was the raw data of the low–range accelerometer, wide–range accelerometer, gyroscope, and magnetometer of the IMU. The output of the first sub–component was the *Short Physical Performance Battery* (SPPB) score and the output of the second sub–component the *Timed Up & Go* (TUG) score the older adult would achieve in one of those standardised geriatric mobility assessments [22, 11]. Chapter 2 features a detailed description of the component and its sub–components.

Component 2 measured the change in body weight as nutrition related physical parameter. An unintended loss in body weight of 5% or more in three month and a *Body Mass Index* (BMI) less than 20 kg/m^2 are indicators for malnutrition. Having a severe impact on health condition and rehabiliation outcomes, malnutrition is an important aspect and of great interest for medical professionals [28]. For that purpose, the component estimated the time spent for preparing meals based on home automation and power consumption sensors installed in the kitchen. The change in body weight was derived from the change in preparation time. Chapter 3 addresses the component in more detail. The third component included

two sub–components. While the first sub–component monitored the activity in the domestic environment of older adults, the second identified behaviour anomalies on daily scale. Regarding the activity monitoring, the flat was modelled as a mathematical weighted directed graph and the weights were the transition between rooms. The cohort was split into an intervention group and a control group. The participants of the intervention group were following the OTAGO exercise programme. The activity of the control group and the intervention group were analysed and the trend was used to control the sensitivity of the third component. The sub–component and the analysis are explained in depth in Chapter 4. The second sub–component used home automation and power consumption sensor data for creating one day activitiy probability maps, i.e. computes the probability of a sensor event occuring in a certain hour of the day. Then, the activity probability map was compared to a baseline and if the deviation was larger than a given threshold the day was classified as abnormal and normal otherwise. The baseline was updated with the normal days to adapt to the behaviour of the older adult. A detailed description of the sub–component can be found in Chapter 5.

In Chapter 6 all components are integrated into a prototype. Implications and future work are shortly discussed as well and further extended in Chapter 7. The foundation of the system and its development was the dataset collected during the 10–months observational randomised pilot intervention study OTAGO conducted by the University of Oldenburg from July 2014 to June 2015. The aim of the study was to investigate the "*Effect of the fall prevention intervention program OTAGO on older persons with frailty and prefrailty—An analysis with conventional geriatric and technology based assessments*" [27]. Twenty (pre–)frail (Frailty Index ≥ 1) older adults (17 female, 3 male), aveage aged ≥ 84.8y (Standard Deviation (SD) ± 5.2y), participated, with an average participation time of 36.5 weeks, in the study. Two participants deceased during the study and hence dropped out before the end. A comprehensive set of physical, psychological, and body parameters were assessed every 31.3d (SD: 5.3d) by professional staff from the University of Oldenburg; additionally, a multi–component sensor system was installed in the domestic environments of the participants, an IMU, and a Global Positioning System (GPS) receiver were given to them. The set of physical parameters contained the SPPB, TUG, *Hand Grip Strength* (HGS), and body weight, amongst others. In terms of the multi–component sensor system, two different type of sensors—ambient sensors and wearable sensors—were used. The set of home automation sensors contained motion sensors, door contact sensors, and power sensors. Additionally, a four–key switch was installed at the front door and a shock sensor was placed in the bed. Because the participant

could choose the sensors installed in the flat the sensor setup differed from participant to participant. As different the sensor setups were, as different were the participants themselves. Physical and mental particularities of older adults and often multimorbid geriatric patients makes them unique; consequently, finding similarities and generalisable outcomes is difficult. While the living conditions were similar, the level of dependence and medical conditions were different and one of the two participants, who passed away during the study, suffered from cancer. Several participants had severe medical incidents changing there living conditions and nursing care needs. Due to the diversity of the study cohort, the study design, and the frequent assessments similar datasets were not publicly available, as of today. To overcome the challenges arising with the diversity, the regression models and the VAE were designed and trained for each individual separately.

The OTAGO study was approved by the Internal Revenue Board (IRB) of the University of Oldenburg on the 30.04.2014 (protocol code: Drs.27/2014) and conducted in accordance with the guidelines of the Declaration of Helsinki. Written informed consent was obtained from all subjects involved in the study.

2 A Deep Learning Approach for TUG and SPPB Score Prediction of (Pre–) Frail Older Adults on Real—Life IMU Data

Friedrich B, Lau S, Elgert L, Bauer JM, Hein A. A Deep Learning Approach for TUG and SPPB Score Prediction of (Pre–) Frail Older Adults on RealLife IMU Data. Healthcare. 2021; 9(2):149. https://doi.org/10.3390/healthcare9020149

Abbreviations

AUC	Area Under the Curve
BMI	Body Mass Index
IMU	Inertial Measurement Unit
LSTM	Long-Short-Term-Memory
ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
SD	Standard Deviation
SPPB	Short-Physical-Performance-Battery
TUG	Timed Up & Go

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2.1 Introduction

With the demographic change a lot of challenges arise, particularly in the fields of medicine and healthcare. Older adults need frequent medical attention for maintaining their health and physical performance. Functional decline and mobility impairments are symptoms of impending diseases and also for frailty [1]. Early diagnosed diseases can be treated well, impairments can be minimised and there is a good chance of rehabilitation. Additionally, older adults have a high risk of falling and the medical implications can be more serious than the incident itself. Usually, standardised geriatrics assessments are used for monitoring the physical performance and estimating the risk of falling of older adults. Although digital solutions are available, the conventional assessments are still mainly performed under the supervision of a medical professional, therefore it is time consuming in clinical settings. Long-term monitoring on a regular basis assessments exceed the logistic capacities of medical professionals. Another disadvantage of assessments is the test situation itself, because people tend to give their best effort in test situations and studies showed that capacity is not performance [2–4]. Sensor assisted monitoring of the mobility and the physical performance in everyday life situations can give valuable information to health professionals for diagnosis and therapy. IMU sensors are low-cost and unobtrusive sensors for measuring the body's specific force and angular acceleration. The light weight and the small size makes those sensors ideal for carrying around in a pocket or attached to a belt in daily life. Moreover, IMUs are not dependent on external infrastructure like satellites and can be used inside as well as outside without any loss of accuracy.

Machine learning approaches and recently deep learning showed good result in estimating gait parameters and fall risk of older adults [5]. One advantage of deep learning approaches is that the algorithms can approximate arbitrary functions, extract features automatically and the time consuming and difficult step of handcrafting features is not needed [6–9]. This advantage comes at the price of computationally expensive training and hyperparameter optimisation.

In this contribution a machine learning model for predicting the assessment scores for the Short-Physical-Performance-Battery (SPPB) and the Timed Up & Go test (TUG) on IMU data is introduced. The model learned from IMU data collected in everyday life situations from a cohort of older adults (84.75 y, 5.19 y SD). This article is structured as follows, in Section 2.2 the state of the art of mobility assessments and technology approaches for assisting are described. Technology assistance is divided in the two subgroups technology assisted assessments and unsupervised assessment approaches. In Section 2.3 the study for data acquisition, steps of preprocessing the data and the machine learning approach

are explained. The results are shown in Section 2.4 and discussed in the following Section 2.5. In the last section conclusions are drawn and further steps are briefly mentioned.

2.2 State of the Art

Assessing the mobility of older adults is a common task in geriatrics medicine, validated and well accepted assessments like the SPPB [10] and the TUG [11] test are commonly used.

The TUG assesses the mobility of an older adult by getting up from a standardised chair, walking a distance of three metres, turning around and getting back to the chair to sit down. Assistive devices used for walking are permitted but it must be documented and used for any re-tests. The time from the start command "Go" until the patient's buttocks touches the seat again is measured in seconds. Assessment categories are <10 s = normal, no mobility impairment, 11–19 s = minor mobility impairment not relevant in everyday life, 20–29 s = mobility impairment, >30 s severe mobility impairment, need for intervention [11]. Each category has numeric score, no mobility impairment is 1, minor mobility impairment is 2, mobility impairment is score 3, and severe mobility impairment is 4.

The SPPB consists of three parts assessing balance, gait speed and lower limb strength. During the balance test, the participant stands with the feet side by side, in semi-tandem stance and tandem stance for ten seconds each. Habitual gait speed is measured over a distance of 2.40 m, 3 m or 4 m. The chair rise test assess the muscle strength of the lower extremities by measuring the seconds the participant needs to perform five times from sit to stand. A maximum of 4 points for each task can be achieved. A total SPPB score $\leq$9 points was found as cut-off value for fit and frail people [10, 12]. The approaches to support the assessments using technology can be categorised in technology assisted assessments and unsupervised assessments in real-life.

The technology assisted assessments are still performed in a way the assessment is supposed to be in order to use the validated point values for evaluation. Technical devices are used to enhance the measurements and to support the supervisors as well as the participants. In [13] an approach to enhance the TUG test was introduced. The measurements of the values' score were computed and automatically measured using an IMU sensor. The computed values showed a high

correlation to the gold standard measurement under supervision of a health professional using a stopwatch. A system for automated SPPB assessment executions has been developed in [14].

The unsupervised assessment approaches are trying to detect motion combinations from assessments in real-life by sensors. Once a motion combination is recognised the recorded sensor data is used to compute the performance. In [15] an approach for automatically detect TUG execution using an IMU sensor is shown and the results were compared to traditional stopwatch measurements. The TUG could be recognised with an accuracy of 96% and the results showed a strong correlation with the conventional method. As an item of the SPPB test and important parameter for functional decline, gait speed is focused as well. Approaches using cameras to measure the gait speed in domestic environments are introduced in [16–18]. Instead of cameras, more privacy respecting ambient sensors can be used as well like in [19–22]. However, all of those sensors are fixed and measure the gait speed in a single location only. To overcome this disadvantage small portable low-cost IMU sensors can be used to measure gait speed as well. The research in [23, 24] showed, that gait speed estimations based on IMU data are comparable to gold-standard measurements of a GAITRite walkway. The experiments described in [25] showed the validity and reproducibility of using IMU sensors for gait parameter estimation. Besides the gait parameter measurements IMU data can be used to measure the intensity of activities [26].

Deep learning approaches showed good results on estimating gait parameters and the risk of falling on wearable sensor data [5]. Yu et al. used IMU data of Parkinson's disease patients collected during TUG assessments for estimating the fall risk and the severity of Parkinson's disease symptoms. The Convolutional Neural Network achieved an F-measure of 94% for estimating the fall risk and a Root-Mean-Squared-Error of 0.06 for severity estimation [27]. Similar approaches showed good results for people suffering from neurological disorders and multiple sclerosis. The networks achieved an accuracy of 92.1% and an Area Under the Curve (AUC) of 0.88 respectively [28, 29]. Considering the parameters age and gender in addition to the raw sensor data significantly improves the performance of deep learning models [30]. Convolutional Neural Networks are also able to predict the frailty and cognitive dysfunction in respect to the mini-mental state examination of older adults. The network was trained with spectograms of walking-inplace data collected by IMUs. The network achieved an accuracy of 94.63% and 97.59% for frailty and cognitive dysfunction respectively [31]. An Artificial Neural Network in combination with a pressure sensor was used to detect abnormal foot postures. The pressure sensors were placed inside shoes and the Artificial Neural Network classified abnormal foot postures based on the gait

characteristics with an accuracy of 99% [32]. Gait abnormalities can be detected using wrist-worn IMUs as well. A deep neural network trained IMU data collected by smartwatches achieved an accuracy of 88.90%, a sensitivity of 90.60%, and a specificity of 86.20% [33]. Deep Neural Networks also has been used to detect the fall incidents themselves. Using wearable sensor data an accuracy of 97.16% was achieved on the task of fall detection [34]. Using accelerometer data only a sensitivity of 88.20% and a specificity of 96.40% could be achieved in [35].

The mentioned approaches focus on enhancing the execution of geriatrics assessments using technology or detecting motion patterns from assessments in real-life. The approach in the article at hand is different, because the score of an assessment is predicted on IMU data. The participant neither has to execute any special motion patterns nor to complete the assessment in a certain place.

2.3 Methods and Materials

Data Acquisition

The data was collected during the observational OTAGO study in 2014 and 2015 over a period of 10 months [36]. The functional performance was assessed every month by the TUG and SPPB tests through conventional stopwatch measurements by a health professional. The cohort consisted of 20 participants (17 female, 3 male) aged 76 to 92 (mean 84.3 y, SD 5.19 y). At baseline, 14 participants (70%) were identified as frail (FrailtyIndex $\geq$ 2 pts.) and six participants were pre-frail. The mean scores of the functional performance were 17.9 s for TUG and 5.95 points for SPPB. The baseline characteristics are presented in Table 2.1. Table 2.2 shows the characteristics of the cohort at the end of the study. The cohort size is reduced by two, because two participants deceased during the study. Despite dropout the available data of these two participants was considered. Each participant got one IMU sensor for data collection.

The IMU sensor of type Shimmer 3r was capable of measuring force in 9 Degrees of Freedom (9DOF) and was comprised of wide range and low range accelerometer, gyroscope, magnetometer, and pressure sensor [37]. In the first two weeks of the study the sensor was set to 51.2 Hz and in the remaining time of the study the sensor was set to 102.4 Hz. The sensors were given to the participants before the TUG and SPPB were done and collected after around about two weeks. Therefore, the dataset consists test situations and everyday life situations. Participants were asked to wear the the Shimmer3r IMU the whole day on a sensor belt, in a trouser pocket or other small pocket on the right side of

Table 2.1 The baseline characteristics of the study cohort

$n = 20$	Age (y)	BMI ($\frac{\text{Kg}}{\text{m}^2}$)	Frailty Index (pts.)	SPPB (pts.)	TUG (s)
Mean	84.75	27.39	1.90	5.95	17.87
SD (±)	5.19	6.10	0.72	2.33	5.33
Range (min-max)	76.00–92.00	17.33–43.09	1.00–3.00	3.00–11.00	11.16–31.63

Table 2.2 The characteristics at the end of the study cohort

$n = 18$	Age (y)	BMI ($\frac{\text{Kg}}{\text{m}^2}$)	Frailty Index (pts.)	SPPB (pts.)	TUG (s)
Mean	85.44	28.27	2.00	6.61	16.12
SD (±)	4.92	6.44	0.97	2.85	5.85
Range (min-max)	77.00–93.00	16.89–45.99	0.00–4.00	2.00–12.00	8.15–30.06

the hip. The logo should face the front and the side with the charging port facing down. At night the sensor was supposed to be placed on the charging station. The participants were instructed to store the sensor safely during taking a shower and in the exercise bath. In total, 259 days of IMU data were collected.

Preprocessing

Before using the data for learning several preprocessing steps were applied. Participants 2, 3, and 4 were excluded, because of unavailable IMU data. The models are tested with two different testing strategies. The first testing strategy is the common testing strategy in machine learning, a test set is separated from the dataset. The second strategy is to test the models on data of participants which have been excluded before processing the data. The data of the excluded participants were neither used for training nor for evaluation of the model during training time. The participants 16 and 19 were randomly chosen for evaluating the performance of the model on unknown participants. This led to an exclusion of the SPPB score 2, because participant 19 was the only participant with a score of 2 in a SPPB assessment. Only two TUG assessments took longer than 30 s (max. +1.6 s). The scores were included in score 3, for not losing the data. The values of the low range accelerometer, wide range accelerometer, gyroscope and

magnetometer were chosen as input to the network. The values of the temperature sensor were mostly 0 and contained errors and were not considered. The orientation in relation to the earth coordinate frame of the IMU was unknown and to eliminate the influence of the orientation the magnitude for each sensor modality was computed by

$$m_i = \sqrt{x_i^2 + y_i^2 + z_i^2} \tag{2.1}$$

where i is the index of the value and x, y, and z are the axis of the sensor. After computing the magnitude the sensor data was filtered by a second order low-pass filter with a cut-off frequency of $\frac{1}{4} \times$ *sampling frequency*. The filtered data has been divided into 5 s non-overlapping windows. Since two different sampling frequencies were used in the study the number of values per window differed. The number of values per window were rounded down to 500 values for the sampling frequency of 102.4 Hz and to 250 values for 51.2 Hz. Latter were oversampled to 500 values per window by duplicating each value. The final sets may contain data from assessments as well, but the amount of windows containing assessment data is insignificant. For each participant about 180,018.13 (SD $\pm$ 19,565.80) samples are available and a maximum of 1320 samples of each participants contain assessment data. The number of windows per class of the resultant dataset are shown in Table 2.3. Due to the setting of the data acquisition the classes were imbalanced and for several classes no windows were available at all. To balance the dataset windows of the overrepresented classes were deleted and the score 12 was excluded from the SPPB. Considering the smallest class as lower threshold for balancing, would have led to massive loss of data. The dataset was balanced two times and two different datasets, one for each assessment were derived. The dataset for the SPPB had 148,307 windows per class and the dataset for the TUG had 169,711 windows per class. The sets were divided into subsets for training (75%), validation (15%) and test (10%) in a stratified fashion.

The data of the two participants reserved for testing were preprocessed in the same way, but were not balanced and not divided into subsets. The data for SPPB score 2 of participant 19 were not considered for evaluation, because the model was not trained for that class. The number of values per class for participants 16 and 19 are shown in Tables 2.4 and 2.5 respectively. The data was collected over a certain period of time and the physical performance changed during that time. So, one participant could have achieved different assessment scores.

Before the data was fed to the model the data was scaled to a range of 0 and 1.

Table 2.3 This table shows the number of windows for each class of the SPPB and the TUG assessments. The range of the TUG score is smaller than the range of the SPPB score

Score	TUG	SPPB
1	216,624	0
2	2,016,236	83,248
3	467,412	128,331
4	–	371,263
5	–	298,456
6	–	235,227
7	–	522,077
8	–	357,449
9	–	379,761
10	–	145,679
11	–	178,402
12	–	379

Table 2.4 This table shows the number of windows for participant 16 who was excluded from the training set

Score	TUG	SPPB
2	88,733	0
3	80	0
4	–	14,505
10	–	43,708

Table 2.5 This table shows the number of windows for participant 19 who was excluded from the training set. Score 2 of the SPPB were not considered for evaluation, because the model for the SPPB was not trained with class 2

Score	TUG	SPPB
2	28,336	256
3	82,010	8633
4	–	28,184
5	–	67,403
7	–	5868

Network Architecture

For this research a deep neural network approach was used. The architecture is shown in Figure 2.1 and the blocks are shown more detailed in Figure 2.2. The architecture and window size are adapted from [38]. Two models were trained,

one for each assessment. The difference of the models was the number of neurons of the classification layer. For the SPPB model 9 neurons, and for the TUG model 3 neurons were used. In both networks the final classification layer is activated by the softmax function. The trained networks are available in the supplementary materials.

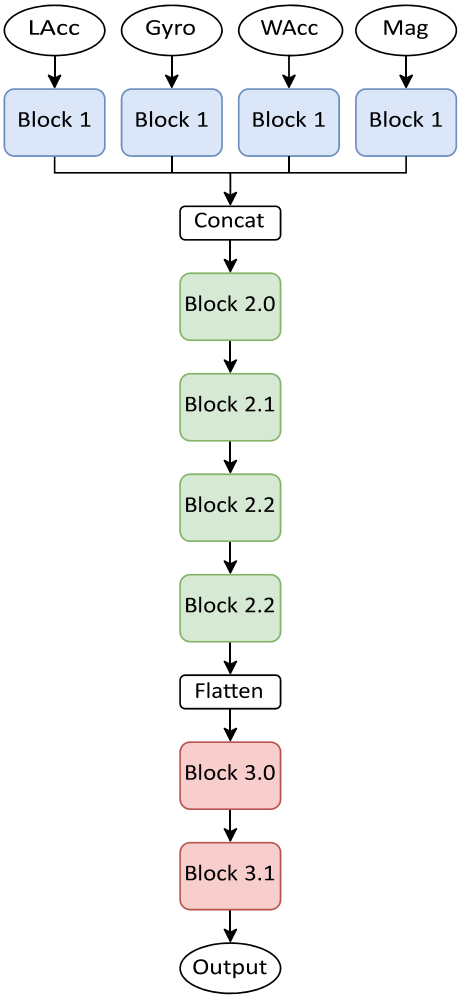


Figure 2.1 The deep neural network used for this research. Each sensor modality had its own input. The inner structure of the blocks are shown in Figure 2.2

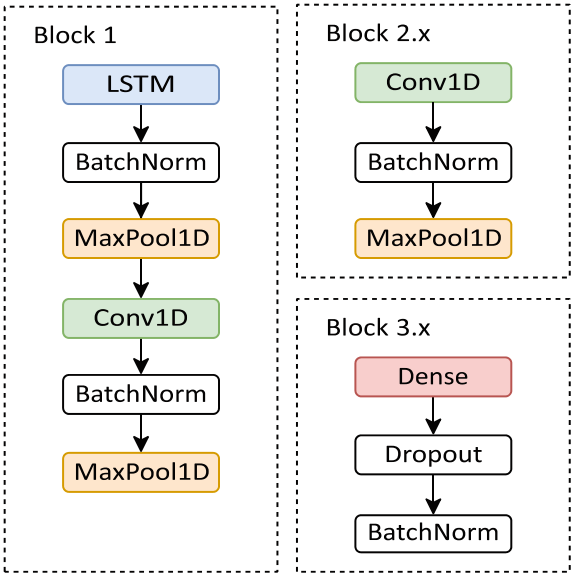


Figure 2.2 The blocks of the deep neural network. Details about the blocks and the layer parameters can be found in the Tables A1–A6 in Appendix A

As first layers Long-Short-Term-Memory (LSTM) layers are used for capturing the relation between the time steps. Then a convolutional layer was added to learn features for each sensor modality. All intermediate features were concatenated and forwarded to a sequence of convolutional layers. The final classification was performed by a two layer neural network. Dropout, maximum pooling and batch normalisation layers were added to prevent overfitting.

The model was trained using categorical cross-entropy as loss function and accuracy as metric. The accuracy was computed as follows

$$\text{accuracy} = \frac{\text{correct classified}}{\text{all samples}} \tag{2.2}$$

The optimiser was AMSGrad version of the *Adaptive Moment Estimation* (Adam) with an initial learning rate of 0.001, a first order derivative momentum of 0.5, a second order derivative momentum of 0.8, and an exponential decay after the first 10 epochs [39].

$$lr(epoch) = 0.001 \times e^{0.1 \times (10 - epoch)} \tag{2.3}$$

2.4 Results

The best epochs were epoch 67 for the SPPB model with a validation accuracy of 94.29% and epoch 52 for the TUG model with a validation accuracy of 95.89%. The accuracy on the test set was 94.28%, and 95.79% for the SPPB and the TUG model respectively. The accuracy for the TUG scores (2, 3) for participant 16 was 98.84%, and 26.15% for participant 19. The accuracy for the SPPB scores (4 and 10) of participant 16 was 6.39%, and for SPPB scores (3, 4, 5, and 7) for participant 19 was 14.13%. The Tables 2.6–2.8 give an overview over the results. The Receiver Operating Characteristic (ROC) curves in Figures 2.3 and 2.4 show high true positive and low false positive rates at high decision thresholds. The average ROC curves (blue) and the ROC curves for each class are showing similar progress and overlap. The AUCs of the TUG model and scores are 0.99 and the AUCs of the SPPB model are 1 except for the classes 7 and 9, where the AUCs are 0.99. The confusion matrices for the SPPB and TUG models are shown in Tables 2.9 and 2.10. The SPPB model classified score 11 best with 573 false classifications (sensitivity: 96.14%, specificity: 99.27%) and score 9 worst with 1381 false classifications (sensitivity: 90.69%, specificity: 98.85%). The TUG model classified score 3 best with 771 (sensitivity: 96.44%, specificity: 97.93%) false classifications and score 2 worst with 1082 false classifications (sensitivity: 95.01%, specificity: 97.25%). The most false classification of the TUG model is for adjacent classes, e.g., 614 samples of class 1 were classified as class 2, but only 268 samples as class 3.

The Figures 2.5 and 2.6 are showing the progress of the loss during training and the Figures 2.7 and 2.8 are showing the accuracy during training. The validation loss and the validation accuracy are fluctuating in the beginning, but become stabilised after epoch 25. Overall, the loss and accuracy graphs showed the desired behaviour, increasing fast in the early epochs and stabilising during the later epochs. The graphs for both models are similar, but the SPPB model shows a little less performance and little higher loss than the TUG model.

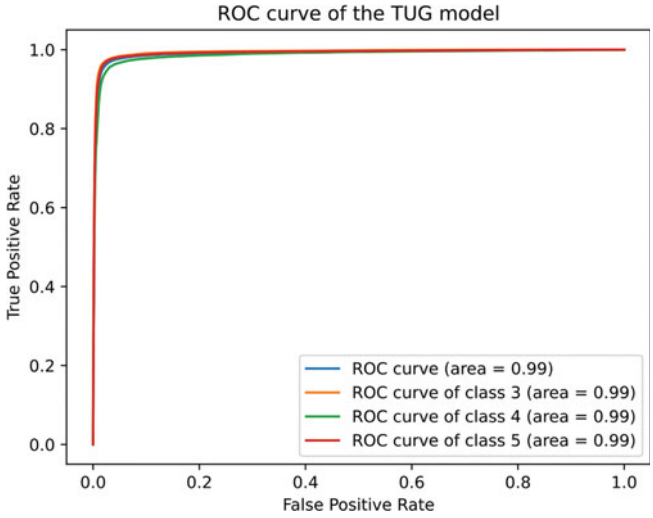


Figure 2.3 The ROC curve of the TUG model

The confusion matrices for participant 16 and 19 are shown in Tables 2.11–2.14, respectively. The accuracy of the model is lower for the unknown participants. The accuracy for the class 2 of the TUG assessment is 98.93% and 99.49% for participant 16 and 19 respectively. The accuracy for the class 3 of the TUG assessment is 2.50% and 0.81% for participant 16 and 19 respectively. The class 1 is not available for the two participants. The best class for the SPPB score of participant 16 is class 4 with an accuracy of 12.51% and the best class for participant 19 is class 6 with an accuracy of 17.23%. The worst classes are 10 for participant 16 and 3 for participant 19 with an accuracy of 0.000641% and 0.00% respectively.

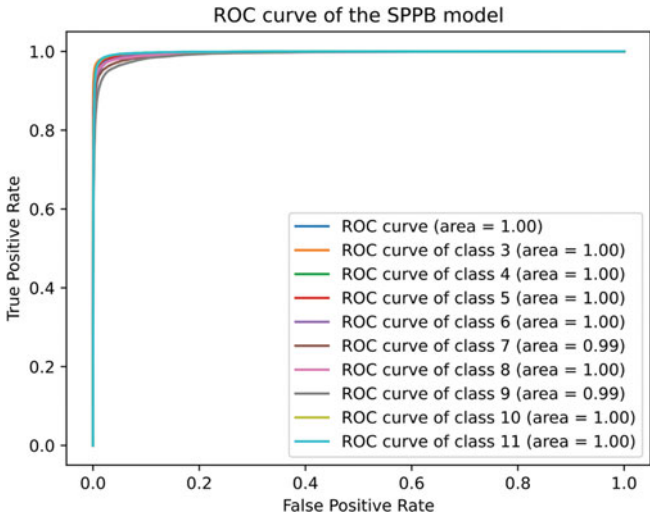


Figure 2.4 The ROC curve of the SPPB model

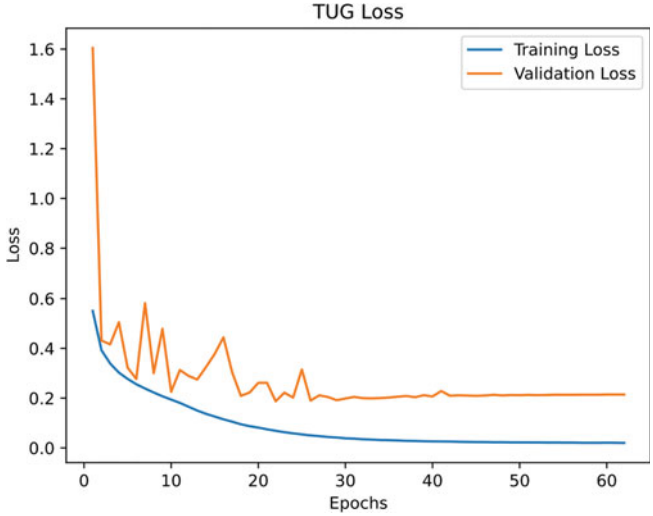


Figure 2.5 The loss of the TUG model. For the first 25 epochs the loss indicates that the learning rate is slightly too large. From epoch 25 the progress shows an asymptotic behaviour

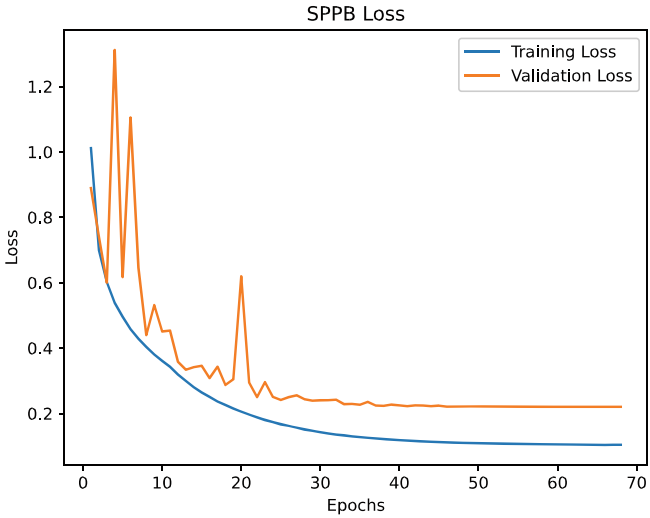


Figure 2.6 The loss of the SPPB model. For the first 25 epochs the loss indicates that the learning rate is slightly too large. From epoch 25 the progress shows an asymptotic behaviour

Participant 16 (76 y) was undergoing chemotherapy two months after the study began and deceased two months before the study ended. Therefore, values of only three assessment dates were available. Facing a severe loss of physical condition following inactivity, functional performance decreased very fast—especially in gait speed, but not in total scores of the SPPB. The interesting part is that, the intra-individual range varied from 4 to 10 points within the short period of time. TUG stopwatch measurements showed a more linear decline and a category change from 2 to 3.

The TUG results for participant 19 (90 y) were very close to the decision boundary of classes 2 and 3. The time of 3 of 6 TUG assessments were of an average of 0.39 s slower than the maximum of 19 s needed for scoring 2 points. The participant scored 3 in those assessments.

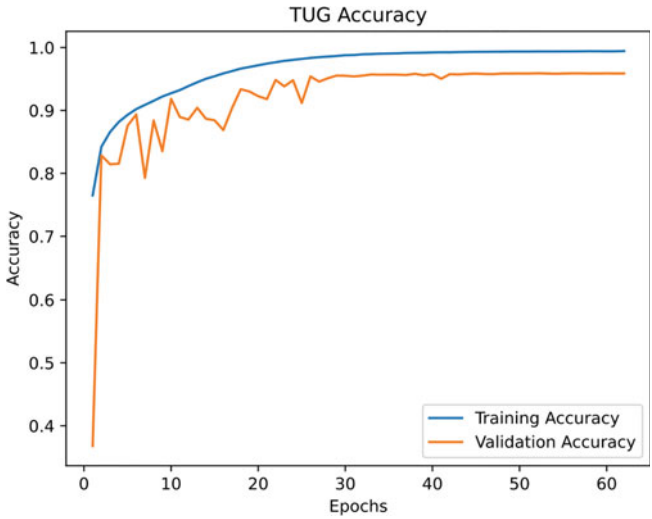


Figure 2.7 The accuracy of the TUG model. According to the progress of the loss, the accuracy fluctuates in the first 25 epochs and shows an asymptotic behaviour after epoch 25. The best validation accuracy score was 95.89% at epoch 52

Limitations

The cohort does not represent the complete scale of the SPPB and the scores 1, 2, and 12 are not learned by the model and hence the validated assessment is not completely represented by the model. Another point is the IMU, which was used by the participants independently. Using filtering approaches some invalid data was filtered, but certainly not all, e.g., if a participant would have given the IMU to another person, the invalid data of that person remains in the dataset. Inactivity covered by noise in the sensor signal will also remain in the dataset.

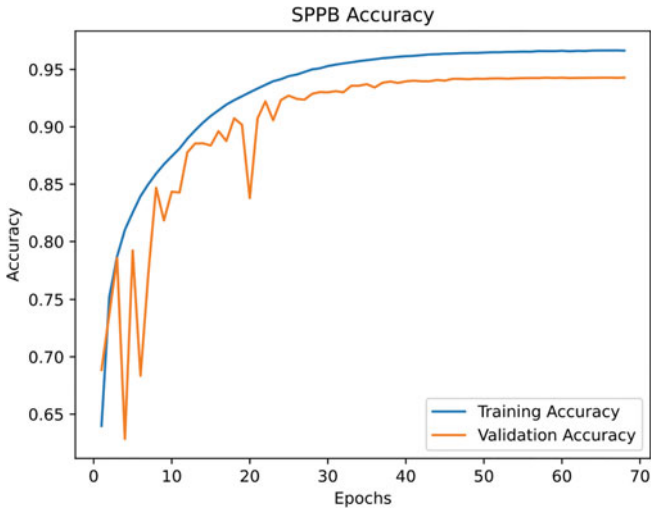


Figure 2.8 The accuracy of the SPPB model. According to the progress of the loss, the accuracy fluctuates in the first 25 epochs and shows an asymptotic behaviour after epoch 25. The best validation accuracy score was 94.29% at epoch 67

Table 2.6 The accuracy on the test set and the two excluded participants

Assessment	Test Set	P16	P19
SPPB	94.27%	6.39%	14.13%
TUG	95.79%	98.84%	26.15%

Table 2.7 The specificity and sensitivity for each TUG score on the test set

Score	Sensitivity	Specificity
1	95.93%	98.50%
2	95.01%	97.25%
3	96.44%	97.93%

Table 2.8 The specificity and sensitivity for each considered SPPB score on the test set

Score	Sensitivity	Specificity
3	95.72%	99.67%
4	95.62%	99.38%
5	94.48%	99.45%
6	93.61%	99.06%
7	92.23%	99.25%
8	94.09%	99.31%
9	90.69%	98.85%
10	95.93%	99.31%
11	96.14%	99.27%

Table 2.9 The confusion matrix of the TUG model. The class with the least false classifications is 1 and the class with the most false classifications is 3. t = true label, p = predicted label

t\p	1	2	3
1	20,780	614	268
2	455	20,580	627
3	193	578	20,891

Table 2.10 The confusion matrix of the SPPB model. The class with the least false classifications is 3 and the class with the most false classifications is 9. t = true label, p = predicted label

t\p	3	4	5	6	7	8	9	10	11
3	14,195	98	35	147	80	95	123	47	10
4	39	14,181	44	163	111	40	157	69	26
5	27	67	14,012	205	72	76	187	155	29
6	99	153	136	13,883	107	116	140	107	89
7	61	154	70	118	13,677	123	229	86	312
8	73	43	52	154	121	13,954	188	97	148
9	62	144	169	173	211	189	13,449	221	212
10	13	46	117	88	47	73	180	14,226	40
11	12	25	24	73	136	106	155	42	14,257

Table 2.11 The confusion matrix of the TUG for participant 16. The participant scores 2 and 3, but not 1. The most values are in class 2. Due to the imbalance the accuracy is high, but the performance is low. t = true label, p = predicted label

t\p	1	2	3
1	0	0	0
2	8	87,782	943
3	1	77	2

Table 2.12 The confusion matrix for SPPB of participant 16. The only class with correct predictions is class 4. t = true label, p = predicted label

t\p	3	4	5	6	7	8	9	10	11
3	0	0	0	0	0	0	0	0	0
4	1814	5643	7484	4245	3043	9421	9322	4041	92
5	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0
10	3921	15	2977	814	7112	22,327	4832	28	1682
11	0	0	0	0	0	0	0	0	0

Table 2.13 The confusion matrix for TUG of participant 19. The model is not able to distinguish between score 3 and score 2 for this participant. t = true label, p = predicted label

t\p	1	2	3
1	0	0	0
2	3	28,191	142
3	86	81,259	665

Table 2.14 The confusion matrix for SPPB of participant 19. This participant was the only one with a SPPB score of 2. Since the model was not trained to classify this score, the values for score 2 were not considered for classification. t = true label, p = predicted label

t\p	3	4	5	6	7	8	9	10	11
3	197	1	37	1	13	3	0	1	3
4	39	1,756	11	235	1753	328	403	99	4009
5	28	14	19,027	299	6143	444	1255	778	198
6	2276	219	15,315	974	363	2388	14,474	30,534	860
7	0	0	0	0	0	0	0	0	0
8	8	594	70	267	220	398	4234	75	2
9	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0	0

2.5 Discussion

The results showed that the models were performing well on all classes on known participants, but significantly worse on unknown participants. Finding one patient the network is not performing well for already proofs that the model cannot be used for unknown participants without further adjustments like fine tuning. The reason why the SPPB model performs slightly worse than the TUG model is that assessment has more classes and is more complex in execution than the TUG. Moreover, the score of balancing test of the SPPB is subject to the impression of the supervising professional, rather than an objective measurement. The models learned the variety amongst the participants of the OTAGO study, even though the cohort was very heterogeneous. The heterogeneity is expected to be the reason for the low model performance for the unknown participants 16 and 19. The TUG assessment results of participant 16 were very close to the decision boundary of two classes. This made it even more difficult for the model to distinguish the data of those classes for an unknown participant correctly. Considering Table 2.9, this seems to be a general problem of the TUG model. That shows the limitations of the assessment scores and boundaries. The scores are defined using full seconds, but with today's technology much more precise measurements up to milliseconds are possible. The low accuracy on the SPPB data is mainly due to the exclusion from the training set, i.e. the participant is unknown to the network.

Even though the ROC curves, the AUCs and the accuracy show a high performance of the model, inferring reasons for score changes are not possible. A change in one item of SPPB changes the score of the assessment, so the change could be due to decreasing gait speed, a declining balance or decreasing lower limb strength. The same holds for the TUG test, because the same aspects are implicitly assessed. Standing up from a chair is dependent on the balance and lower limb strength, and the walking part assesses the gait speed. So, the TUG score incorporates the same dimensions like the SPPB.

The results showed that the models were performing well on all classes on known participants, but significantly worse on unknown participants. The loss and accuracy graphs showed the desired behaviour of increasing fast in the early epochs and stabilising during the later epochs. The graphs for both models were similar, but the SPPB model showed a slightly lesser performance and slightly higher loss than the TUG model. Since the SPPB assessment has more classes and is more complex in execution than the TUG, this is a reasonable finding. Moreover, the score of balancing test of the SPPB is subject to the impression of the supervising professional, rather than an objective measurement. The models learned the variety amongst the participants of the OTAGO study, even though the cohort was very heterogeneous. However, this heterogeneity is expected to be the reason for the low model performance for the unknown participants 16 and 19 as well.

Participant 16 (76 y) was undergoing chemotherapy two months after the study began and deceased two months before the study ended. Therefore, values of only three assessment dates were available. Facing a severe loss of physical condition following inactivity, functional performance decreased very fast—especially in gait speed, but not in total scores of the SPPB. The interesting part is that, the intra-individual range varied from 4 to 10 points within the short period of time. TUG stopwatch measurements showed a more linear decline and a category change from 2 to 3.

The TUG results for participant 19 (90 y) were very close to the decision boundary of classes 2 and 3. The time of 3 of 6 TUG assessments were on an average of 0.39 s slower than the maximum of 19 s needed for scoring 2 points. The participant scored 3 in those assessments. This made it even more difficult for the model to distinguish the data of those classes for an unknown participant correctly. The low accuracy on the SPPB data is mainly due to the exclusion from the training set, i.e. the participant is unknown to the network. The results show as well that the age is not important itself. Participant 16 is 14 y younger than participant 19, but the physical condition is much worse. The physical condition of participant 19 is clinically constant and varies slightly between two classes.

The latter shows the limitations of the assessment categories and boundaries. The categories are defined using full seconds, but with today's technology much more precise measurements up to milliseconds are possible.

2.6 Conclusions and Future Work

The results showed that it is possible to use machine learning to predict the geriatrics mobility assessment scores on real-life IMU data, even though the cohort was very heterogeneous and the IMUs were not rigidly attached to the body. The models performed well on known participants and were able to predict the scores of SPPB and TUG with an accuracy of 95.79% and 94.27% for the TUG and SPPB assessment respectively. The ROC curves, the specificities and the sensitvities for each class show, that the models performing well enough to be used by professionals. On the downside, the results showed that the models are very inaccurate on data of unknown participants.

The most promising approach is to use deep unsupervised learning. In the first step the network could be trained to approximate the underlying probability distribution of the training data. In the next step fine tuning for the new participant could be applied. Since one participant is likely to show only a certain subset of all available scores, the output of the network is the probability whether the sample belongs the class the network was fine tuned with. So, changes could be detected.

Another important point is the investigation of the performance on a different cohort. The cohort used for this research was very special, e.g., study inclusion criteria was being pre-frail at least. For healthier cohort of older adults an early detection of physical decline is important for early treatment and prevention as well. Long-term monitoring would be useful for those cohorts. So the models must be evaluated on data of a healthier cohort. Moreover, measures to simplify the model without loss of accuracy should be taken, because the data acquisition is difficult and costly. A less complex model needs less training data.

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3 Detecting Impending Malnutrition of (Pre–) Frail Older Adults in Domestic Smart Home Environments

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Abbreviations

(i)ADL	(instrumental) Activities of Daily Living
BMI	Body Mass Index
BW	Body Weight
ESPEN	European Society for Clinical Nutrition and Metabolism
HGS	Hand Grip Strength
KPI	Key Performance Indicator
MCID	Minimal Clinically Important Difference
N/A	Not Applicable
SD	Standard Deviation
SPPB	Short Physical Performance Battery
TUG	Timed Up & Go

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3.1 Introduction

Malnutrition as a result of insufficient nutritional intake or increased requirement is a problem of international interest and one risk factor for functional decline and loss in performance of older adults [1–3]. As a functional parameter the hand grip strength (HGS) is affected by malnutrition and hence the activities of daily living (ADL) as well [4, 5]. One indicator for malnutrition is the significant loss in body weight. The guidelines on Enteral Nutrition: Geriatrics consider an unintentional loss of more than 5% in 3 months in body weight and a Body Mass Index (BMI) less than 20 kg/m^2 as indicators for malnutrition [4]. A continuous monitoring of the body weight enables physicians to investigate the cause of major fluctuations of the body weight and choose a proper treatment at an early stage of an impending malnutrition. However, a continuous monitoring of the body weight by professionals exceeds their capacities and self-monitoring tends to be unreliable in the target group of older (pre-) frail adults [6]. Low-cost, unobtrusive ambient sensor systems are well-accepted among the group of older adults if installed in a flat for continuous monitoring [7, 8]. The aim of this contribution is to show that the meal preparation time (key performance indicator, KPI) is associated with body weight changes as an indicator for impending malnutrition. To validate the suitability of the KPI, the relation between HGS and KPI was investigated and based on the results the relation between the KPI and the body weight was investigated. We used the data of an observational randomised controlled trial study with initially 20 participants aged 84.75 y with a Standard Deviation (SD) of 5.19 y.

This article is structured as follows. The Section 2 is a short survey of existing approaches to estimate the nutritional intake, measuring the body weight and detecting (instrumental) ADL using information technology. In Section 3 the OTAGO study for collecting the data, the preprocessing of the dataset, and the used algorithms are explained. In Section 4 the results are shown. The discussion of the results can be found in Section 5. In the last section conclusions are drawn from results and discussion.

3.2 State of the Art

This section is divided into two parts. The first part covers the state of the art of monitoring the nutritional intake and the body weight in domestic smart home environments and the second part covers the state of the art of detecting ADL in domestic smart home environments.

Nutritional Intake and Body Weight

A popular approach to monitor the nutritional intake in smart homes is to use smart appliances in the kitchen [9]. As part of a smart kitchen a smart fridge tracks the food and estimates the nutritional intake based on ingredients [10–12]. Moreover, the fridge has the capability to suggest recipes based on the estimated nutritional intake and the ingredients in the fridge.

Another essential appliance for preparing meals is the stove. When used for cooking most of the used ingredients are coming together at the stove and hence it is a suitable place for tracking the used ingredients. The quality of the nutritional intake estimation can be enhanced by communication with other smart appliances such as the smart fridge [13]. If no smart fridge is available a smartphone application for determining the nutritional facts is introduced. Adding a smart weighting scale to the already smart kitchen gives the essential information of the amount of each used ingredient and further refines the estimation [14]. An approach not relying on a smart kitchen uses groceries receipts to analyse the bought food and derive nutritional facts from that information [15].

Using smart body weighting scales is another approach to measure the body weight changes as an indicator of malnutrition. The results are quite accurate compared to inperson measurements [16]. Using ambient sensors for measuring the body weight is an interesting approach as well, because no effort by a person is needed to obtain the values. People are nearly using their beds every day and a daily weight measurement can be obtain by installing load cells under the feet of the bed [17].

The disadvantages of the smart kitchen approach are the costs and the usability. To transform a kitchen into a smart kitchen, renovation is necessary and the rehab in combination with the new appliances likely exceeds the funds of older adults. Moreover, a certain procedure for preparing meals is needed and a smartphone is involved as well. Especially, older adults are encounter difficulties using smartphones or following a certain unknown procedure.

The smart body weighing scale approach is promising, but the measurements are varying and tend to be unreliable for individuals with a body weight of more than 110 kg [16].

Activities of Daily Living

In Geriatrics Medicine distinction is made between activities of daily living (ADL) and instrumental activities of daily living (iADL) [18]. ADLs are daily

self-care activities, like bathing, grooming and homemaking and essential for fulfilling the basic psychological and physiological needs of humans [19]. The iADLs are more complex activities of daily living, like cooking, groceries, and using means of transportation [20]. The ability to perform the (i) ADL is an indicator for independence. A common approach is to use activation sequences of sensors to define sequences and the type of the sensor, motion and power consumption, to determine the (i) ADL [21, 22]. The sequences are used to train machine learning models, e.g., for prediction purposes [23]. If none or only a few ground truth labels are available unsupervised learning is used to detect activities of daily living. Based on the cluster structure all unknown samples can be labeled [24, 25]. For recognition enhancement wearable sensors and smartphones are combined with the ambient sensors [26, 27]. Furthermore, efforts for adding more sensors like temperature, hygrometry and microphones were made for recognising (i) ADL [28].

All these approaches have one thing in common, they are relying on a homogeneous set of sensors and mostly on labeled data. Both is not available in the dataset used for the research in this contribution. Moreover, only two approaches have been tested on real-world data of older adults [22, 23]. None of the approaches are trying to find a relation between the meal preparation and the physiological parameter body weight.

3.3 Materials and Methods

Data Acquisition

The dataset has been collected during the OTAGO study conducted from July 2014 to December 2015 by the Carl von Ossietzky University of Oldenburg. The aim of the study was to validate the effect of the OTAGO exercise programme on (pre-)frail older adults. The participants were recruited from older adults who had previously participated in studies of the university and and from residents of a local sheltered home. The inclusion criteria was at least of all a pre-frail condition (Frailty Index = 2). In total, 20 participants (3 m, 17 f) have been included in the study and have been observed for 40 weeks with monthly measurements of a comprehensive geriatric assessment including Short Physical Performance Battery (SPPB) [29], Timed Up & Go (TUG) [30], Hand Grip Strength (HGS), Frailty Index [31] and instrumental Activities of Daily Living (iADL) [32], and the monitoring of the body weight and BMI, which have been conducted by research staff of the University of Oldenburg. The SPPB and TUG are assessing

the mobility of older adults. The SPPB consists of three items, *stance, 4 m walk,* and *5-times chair rise.* All items are scored from 0 to 4 points and overall 12 points can be achieved. The higher the points, the better the mobility. The first items assesses the balance, the second item the gait speed, and the last item the lower limb strength. The TUG assess similar dimensions, but combined in one item. The persons sits on a chair, rises from the chair, walks 3 m, turns around, walks 3 m back, and sits down again. The time is measured and the faster, the time the better the mobility. The HGS of both hands were measured using a JAMAR hand-held dynamometer (JAMAR, Bolingbrook, IL). The HGS of both hands was measured twice and the highest value was used as result. The Frailty Index and iADL are measures for independence. The Frailty Index was assessed by a questionary containing the items *weight loss, exhaustion, physical activity, walk time,* and *grip strength.* Each item is scored from 0 to 1 point, where 1 point denotes impairment. A score of 2 indicates pre-frail condition and a people with a score of 3 or higher are considered as frail. The The iADL questionary contains the eight items *ability to use telephone, shopping, food preparation, housekeeping, laundry, mode of transportation, responsibility for own medications*, and *ability to handle finances.* The items are scored with 0 or 1 points, where 0 is the lowest functional level and 1 the highest. The participants did not report their weight, because self–reporting is less reliable than measurements by professionals. Participants may not pay attention to measure their weigth at the same time, wearing the same apparel. Moreover, a diary containing various information like the cognitive state of health, the recent activities, special events, and visitors was kept by the research staff.

During the study a multi-component sensor system comprised of power and home automation sensors were installed in the flats of the participants. The home automation sensors were motion sensors installed in the rooms and door sensors installed at the main doors and the fridge. The sensors were connected wirelessly to a base station. The flats of the participants had different layouts and a different set of installed sensors, because the participants were free to choose the sensors. A table showing the available sensors for this research for each participant can be found in the appendix. Figure 3.1 shows two different flats of participants. The Tables 3.1–3.4 show the values at baseline, end, and of the subcohort considered for the analysis respectively. Figure 3.2 features the progress of the body weight of the used subcohort over the whole study.

We used the The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines to ensure all important information were included in the article [33]. The checklist can be found in the supplemental materials.

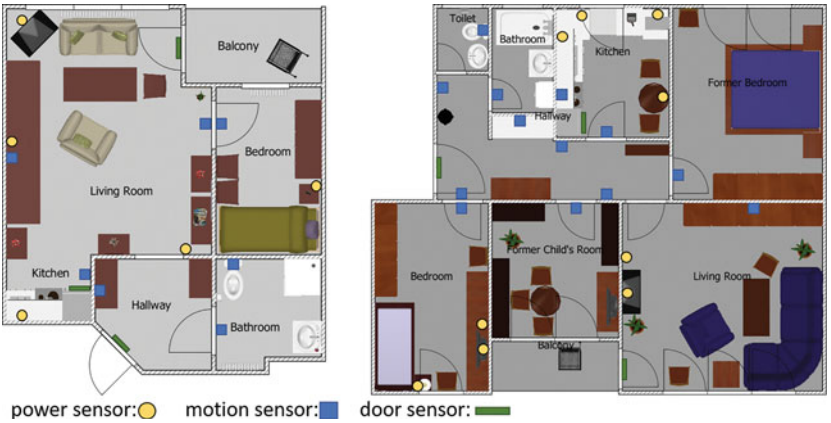


Figure 3.1 The flats of two participants of the OTAGO study. The layout and size of the flats are different and so are the installed sensors. In the right flat a great variety of sensors were installed, especially in the kitchen and the hallway

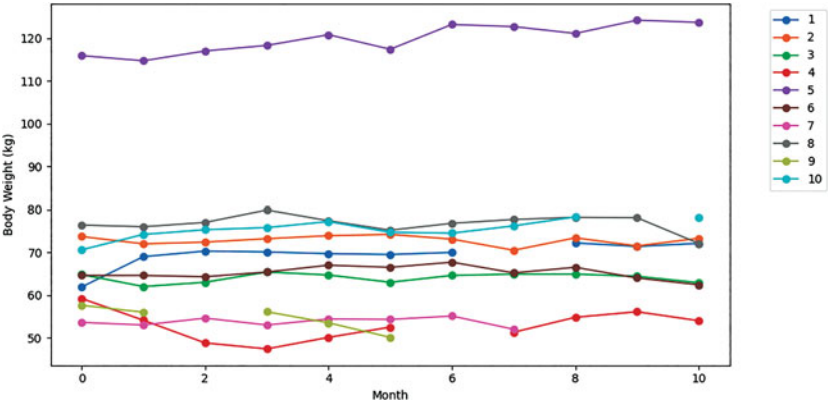


Figure 3.2 The progress of the body weight of the used subcohort over the whole study duration. Gaps in the graphs indicate missing values. The numbers in the legend are the participant IDs

Table 3.1 The characteristics of the study cohort at baseline (T0)

n = 20, m = 3, f = 17	Age (*y*)	Body Weight (kg) *m/f*	HGS (kg) *m/f*
Mean	84.8	66.2/69.2	17.1/14.1
Range (min–max)SD (±)	5.2 76.0–92.0	3.6/17.3 61.9/43.8–70.8/115.9	5.1/7.0 11.7/3.7–24.0/33.0

n = 20, m = 3, f = 17	Frailty Index (pts.)	SPPB (pts.)	TUG (s)	iADL (pts.)
Mean	1.9	6.0	17.9	7.3
Range (min–max)SD (±)	0.7 1.0–3.0	2.3 3.0–11.0	5.3 11.2–31.6	1.4 3.0–8.0

SD: Standard Deviation, HGS: Hand Grip Strength, SPPB: Short Physical Performance Battery, TUG: Timed Up & Go, iADL: instrumental Activities of Daily Living.

Table 3.2 The characteristics of the study cohort at the end (T10)

n = 18, m = 3, f = 15	Age (*y*)	Body Weight (kg) *m/f*	HGS (kg) *m/f*
Mean	84.5	66.1/71.0	16.8/12.2
Range (min–max)SD (±)	4.9 77.0–93.0	5.4/19.1 59.1/42.7–72.1/123.7	4.2/3.7 13.3/5.0–22.7/18.0

n = 18, m = 3, f = 15	Frailty Index (pts.)	SPPB (pts.)	TUG (s)	iADL (pts.)
Mean	2.0	6.6	16.4	6.1
Range (min–max)SD (±)	1.0 0.0–4.0	2.9 2.0–12.0	6.0 8.5–30.06	2.3 1.0–8.0

SD: Standard Deviation, HGS: Hand Grip Strength, SPPB: Short Physical Performance Battery, TUG: Timed Up & Go, iADL: instrumental Activities of Daily Living.

Table 3.3 The characteristics of the used subcohort at baseline (T0)

n = 10, m = 1, f = 9	Age (*y*)	Body Weight (kg) *m/f*	HGS (kg) *m/f*
Mean	85.1	61.9/70.7	24.0/15.4
Range (min–max)SD (±)	4.6 77.0–91.0	0.0/18.6 61.9/53.6–61.9/115.9	0.0/6.1 24.0/7.3–24.0/23.7

n = 10, m = 1, f = 9	Frailty Index (pts.)	SPPB (pts.)	TUG (s)	iADL (pts.)
Mean	1.8	5.7	18.3	7.8
Range (min–max)SD (±)	1.0 1.0–3.0	2.1 3.0–9.1	6.1 12.0–31.6	0.6 6.0–8.0

SD: Standard Deviation, HGS: Hand Grip Strength, SPPB: Short Physical Performance Battery, TUG: Timed Up & Go, iADL: instrumental Activities of Daily Living.

Table 3.4 The characteristics of the used subcohort at the end (T10)

n = 10, m = 1, f = 9	Age (*y*)	Body Weight (kg) *m/f*	HGS (kg) *m/f*
Mean	85.1	72.1/74.8	22.7/14.5
Range (min–max)SD (±)	4.5 77.0–91.0	0.0/21.2 72.1/54.0–72.1/123.7	0.0/5.0 22.7/8.0–22.7/22.7

n = 10, m = 1, f = 9	Frailty Index (pts.)	SPPB (pts.)	TUG (s)	iADL (pts.)
Mean	2.3	5.9	15.5	7.1
Range (min–max)SD (±)	1.1 1.0–4.0	2.5 3.0–10.0	5.3 8.5–22.3	1.6 4.0–8.0

SD: Standard Deviation, HGS: Hand Grip Strength, SPPB: Short Physical Performance Battery, TUG: Timed Up & Go, iADL: instrumental Activities of Daily Living.

Data Preprocessing

Before the preprocessing on data level three exclusion criteria were applied. Two on participant level and one on data level. The first criteria was the *self supporter* criteria. All participants who were not preparing their meals themselves were excluded as well as participants who have been assisted by another person. Accordingly, 2 participants were excluded, because they received meals on wheels, 6 participants, because they were always eating out and 2 participants, because they had assistance. Two participants were partly excluded, because their condition changed from *self supporters* to being assisted. Comparing the iADL questionnaires for the excluded month and participants confirmed their status as none self supporters. The second criteria was *weight change.* If there was no weight change, the month was excluded. A change of 1% or less was considered as no change, because the measurements might have not been performed at the same time and with the same apparel. For all remaining participants a weight change was recorded and 4 month of 4 participants, 1 month for each participant, were excluded. On data level *faulty and unreliable* data has been excluded. Reasons for faulty data are malfunctioning of the sensor base station, failure of the sensors themselves or removed and rearranged sensors by the participant. Applying the criteria 5 months were excluded. After applying all exclusion criteria 10 participants with 7.2 month (SD 2.9) were considered. The group of the excluded participants is too heterogeneous to serve as control group.

The power consumption sensors were measuring the standby power consumption of the appliances and hence the standing by time must be filtered. Using the preprocessed dataset, the time spent for preparing meals was estimated each day.

Meal Preparation Time Estimation

For estimating the time, i.e., key performance indicator (KPI), all sensors in relation to the kitchen were used. Since the sensor setup was different in each flat, the number of sensors was used for computing the time and not the used appliance. In the left flat in Figure 3.1 three sensors were installed in the kitchen, one motion sensor, one door contact sensor in the fridge and one power consumption sensor at the kettle. In the right flat two motion sensors, one door contact sensor in fridge and four power consumption sensors at the microwave, kettle, coffee maker, and lamp were installed. The beginning of preparing a meal has been defined by a motion or door contact sensor event followed by events from at least half of the sensors in the kitchen. When there was no sensor event in the kitchen for 60 min

the preparation was considered as finish. The time was computed for every day of the month. After computing the time values the function

$$f(t) = a_0 + a_1 t \tag{3.1}$$

was minimised using the method of least squares

$$\sum_{i=1}^{n} (f(t_i, a_0, a_1) - t_i)^2 \tag{3.2}$$

where n is the number of days, t_i the computed time for day i and, a_0 and a_1 the parameters of the linear function. The parameters a_0 and a_1 are adjusted according to the minimisation criteria

$$\min_{a_0, a_1} \| f - t \|_2^2 \tag{3.3}$$

Minimal Clinically Important Difference

The minimal clinically important difference (MCID) is the smallest change in a parameter which is identified as important to the patient [34, 35]. The considered parameter and the smallest change are not universally valid, i.e., there are different parameters and a different MCIDs. The MCID for the change in body weight in relation to malnutrition was defined in the ESPEN guidelines and is 5% in 3 months [4].

Statistical Analysis

For assessing the relationship the rank correlation coefficient Spearman's ρ is used. Spearman's ρ assess a monotonic relationships between two variables. In the following $\rho(X,Y)$ denotes the correlation coefficient between the variables X and Y. All correlations with a p-value smaller than 0.001 are considered as significant. We used the *Scientific computing tools for Python (SciPy)* (v1.4.1) for the statistical analysis [36].

3.4 Results

The characteristics of the cohort and the sub-cohort used for this research are shown in Tables 3.1–3.4. The results of our analysis' are shown in the Tables 3.5–3.14. Each table contains the results of one participant and shows the correlation between KPI and body weight ($\rho(KPI, Body\ Weight)$), if MCID, if the participant gained or lost weight (*BW change*), the correlation between the KPI and HGS ($\rho(KPI, HGS)$), and if HGS increased or decreased (*HGS change*). All correlations with the body weight and between the HGS and the body weight and between the KPI and the body weight were greater or equal $\rho = |0.99|$ and significant with $p < 0.001$. The second to the fourth columns of the tables show the correlation between the body weight and the KPI, whether the month is part of a MCID change in body weight and the direction of the change, i.e., gain, loss, no change. The columns five and six show the correlation between the hand grip strength (HGS) and the KPI and the direction as well. The average time spent per day for preparing meals was 10.62 min (SD 19.57) with a minimum of 0.0 min and a maximum of 107.23 min. Overall, there were 8 months with a change in the body weight greater than 5% for 2 participants (ID 1, 4). In two of these months the participants gained weight, in another two the participants weight did not change, and in 4 months the participants lost weight. In 26 months the participants gained weight (ID 1–10) and in 20 months they (ID 1–10) lost weight. There are 18 months were no weight change was measured for participants 1, 2, 3, 5, 6, 7, 8, and 10 and the correlation coefficient could not be computed, i.e., not applicable (N/A).

For the HGS a decrease was measured for 28 months for all participants and an increase for 24 months for all participants except for 9. In 18 months the HGS analysis was not applicable (N/A).

The Figures 3.3–3.6 are showing example plots of the values for 1 months. Figure 3.3 shows the first month of participant 1 and an increase in body weight (61.9–69.0 kg), a slight decrease in HGS (24.0–23.0 kg), and an increase in the KPI (the sixth month of participant 1. The weight gain was smaller than 1% (69.5–70.0 kg) and≈10–15 min). Figure 3.4 shows hence the month was considered as no change. The HGS changed slightly (22.0–21.7 kg) and the KPI was nearly constant. Figure 3.5 shows the second month of participant 4 with a significant loss in body weight (54.1–48.8 kg), a slight decrease in HGS (12.0–11.3 kg) and a strong increase in the KPI (≈39–76 min). Figure 3.6 shows the third month of participant 4 with a slight loss in body weight (48.8–47.4 kg), a slight decrease in HGS (11.3–11.0 kg), and a strong increase in the KPI (≈36–88 min). This month is still part of a three months interval where a MCID occurs.

Table 3.5 The results of the correlation analysis for participant 1

ID	$\rho(KPI, Body\ Weight)$	MCID	BW Change	$\rho(KPI, HGS)$	HGS Change
1	0.99	no	gain	−0.99	decrease
	−1.00	no	gain	1.00	decrease
	N/A	no	no change	N/A	N/A
	N/A	no	no change	N/A	N/A
	N/A	no	no change	N/A	N/A
	N/A	no	no change	N/A	N/A
	−1.00	no	gain	−1.00	increase
	1.00	no	gain	−1.00	decrease
	−1.00	no	loss	1.00	increase
	N/A	no	no change	N/A	N/A

KPI: Key Performance Indicator, MCID: Minimal Clinically Important Difference, BW: Body Weight, HGS: Hand Grip Strength, N/A: Not Applicable.

Table 3.6 The results of the correlation analysis for participant 2

ID	$\rho(KPI, Body\ Weight)$	MCID	BW Change	$\rho(KPI, HGS)$	HGS Change
2	1.00	no	loss	1.00	decrease
	N/A	no	no change	N/A	N/A
	−1.00	no	gain	1.00	decrease
	N/A	no	no change	N/A	N/A
	N/A	no	no change	N/A	N/A
	1.00	no	loss	1.00	decrease
	−1.00	no	loss	1.00	increase
	1.00	no	gain	1.00	increase
	1.00	no	loss	1.00	decrease
	−1.00	no	gain	−1.00	increase

KPI: Key Performance Indicator, MCID: Minimal Clinically Important Difference, BW: Body Weight, HGS: Hand Grip Strength, N/A: Not Applicable.

Table 3.7 The results of the correlation analysis for participant 3

ID	$\rho(KPI, Body\ Weight)$	MCID	BW Change	$\rho(KPI, HGS)$	HGS Change
3	−0.99	no	loss	−0.99	decrease
	−1.00	no	gain	N/A	no change
	1.00	no	gain	−1.00	decrease
	−1.00	no	loss	1.00	decrease
	1.00	no	loss	1.00	decrease
	1.00	no	gain	1.00	increase
	N/A	no	no change	N/A	N/A
	N/A	no	no change	N/A	N/A
	−1.00	no	loss	1.00	increase

KPI: Key Performance Indicator, MCID: Minimal Clinically Important Difference, BW: Body Weight, HGS: Hand Grip Strength, N/A: Not Applicable.

Table 3.8 The results of the correlation analysis for participant 4

ID	$\rho(KPI, Body\ Weight)$	MCID	BW Change	$\rho(KPI, HGS)$	HGS Change
4	1.00	yes	loss	1.00	decrease
	−1.00	yes	loss	−1.00	decrease
	−1.00	yes	loss	−1.00	decrease
	−1.00	yes	gain	1.00	decrease
	−1.00	yes	gain	−1.00	increase

KPI: Key Performance Indicator, MCID: Minimal Clinically Important Difference, BW: Body Weight, HGS: Hand Grip Strength, N/A: Not Applicable.

Table 3.9 The results of the correlation analysis for participant 5

ID	$\rho(KPI, Body\ Weight)$	MCID	BW Change	$\rho(KPI, HGS)$	HGS Change
5	−1.00	no	gain	−1.00	increase
	1.00	no	gain	−1.00	decrease
	N/A	no	no change	N/A	N/A
	−1.00	no	loss	1.00	increase
	−1.00	no	gain	1.00	decrease

KPI: Key Performance Indicator, MCID: Minimal Clinically Important Difference, BW: Body Weight, HGS: Hand Grip Strength, N/A: Not Applicable.

Table 3.10 The results of the correlation analysis for participant 6

ID	$\rho(KPI, Body\ Weight)$	MCID	BW Change	$\rho(KPI, HGS)$	HGS Change
6	−0.99	no	loss	0.99	increase
	−1.00	no	gain	1.00	decrease
	−1.00	no	gain	−1.00	increase
	−1.00	no	gain	1.00	decrease
	−1.00	no	loss	1.00	increase
	−1.00	no	gain	−1.00	increase
	N/A	no	no change	N/A	N/A
	1.00	no	loss	−1.00	decrease
	−1.00	no	gain	1.00	decrease
	N/A	no	no change	N/A	N/A

KPI: Key Performance Indicator, MCID: Minimal Clinically Important Difference, BW: Body Weight, HGS: Hand Grip Strength, N/A: Not Applicable.

Table 3.11 The results of the correlation analysis for participant 7

ID	$\rho(KPI, Body\ Weight)$	MCID	BW Change	$\rho(KPI, HGS)$	HGS Change
7	0.99	no	loss	0.99	increase
	−1.00	no	gain	−1.00	increase
	1.00	no	loss	1.00	decrease
	1.00	no	gain	1.00	increase
	N/A	no	no change	N/A	N/A
	−1.00	no	loss	1.00	increase

KPI: Key Performance Indicator, MCID: Minimal Clinically Important Difference, BW: Body Weight, HGS: Hand Grip Strength, N/A: Not Applicable.

Table 3.12 The results of the correlation analysis for participant 8

ID	$\rho(KPI, Body\ Weight)$	MCID	BW Change	$\rho(KPI, HGS)$	HGS Change
8	N/A	no	no change	N/A	N/A
	−1.00	no	gain	−1.00	increase
	−1.00	no	gain	−1.00	increase
	−1.00	no	loss	1.00	increase
	−1.00	no	loss	−1.00	decrease
	−1.00	no	gain	−1.00	increasae
	1.00	no	gain	−1.00	decrease
	N/A	yes	no change	N/A	N/A
	N/A	yes	no change	N/A	N/A
	1.00	yes	loss	−1.00	increase

KPI: Key Performance Indicator, MCID: Minimal Clinically Important Difference, BW: Body Weight, HGS: Hand Grip Strength, N/A: Not Applicable.

Table 3.13 The results of the correlation analysis for participant 9

ID	$(KPI\,\rho, Body\ Weight)$	MCID	BW Change	$\rho(KPI, HGS)$	HGS Change
9	0.99	N/A	loss	0.99	decrease

KPI: Key Performance Indicator, MCID: Minimal Clinically Important Difference, BW: Body Weight, HGS: Hand Grip Strength, N/A: Not Applicable.

Table 3.14 The results of the correlation analysis for participant 10

ID	$\rho(KPI, Body\ Weight)$	MCID	BW Change	$\rho(KPI, HGS)$	HGS Change
10	−1.00	no	gain	1.00	decrease
	1.00	no	gain	−1.00	decrease
	N/A	no	no change	N/A	N/A
	−1.00	no	gain	−1.00	increase
	−1.00	no	loss	−1.00	decrease

KPI: Key Performance Indicator, MCID: Minimal Clinically Important Difference, BW: Body Weight, HGS: Hand Grip Strength, N/A: Not Applicable.

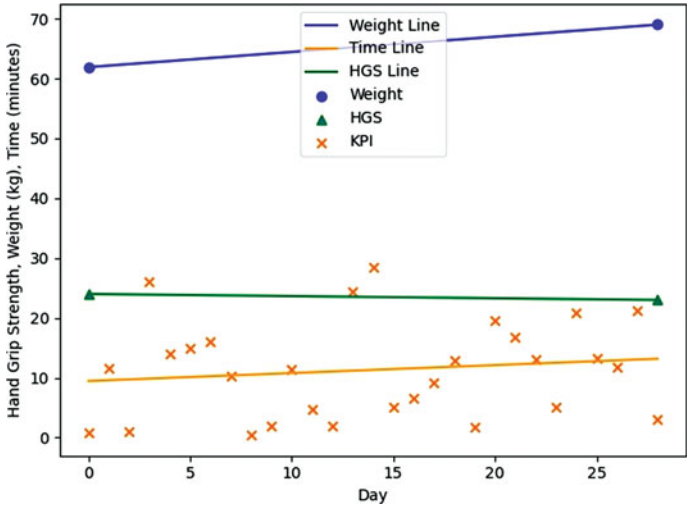


Figure 3.3 The first month of participant 1. Gain in body weight (61.9–69.0 kg), decrease in HGS (24.0– 23.0 kg), increase in the KPI (≈10–15 min). HGS: Hand Grip Strength, KPI: Key Performance Indicator

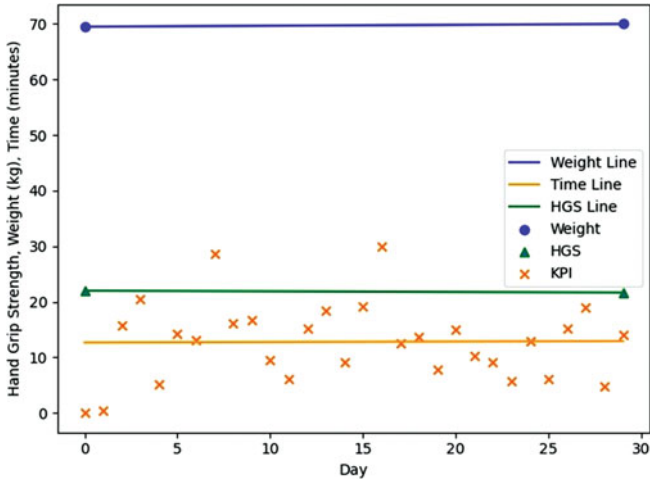


Figure 3.4 The sixth month of participant 1. Gain in body weight smaller than 1% (69.5–70.0 kg) and considered as no change, decrease in HGS (22.0–21.7 kg), nearly constant KPI (≈12–13 min). HGS: Hand Grip Strength, KPI: Key Performance Indicator

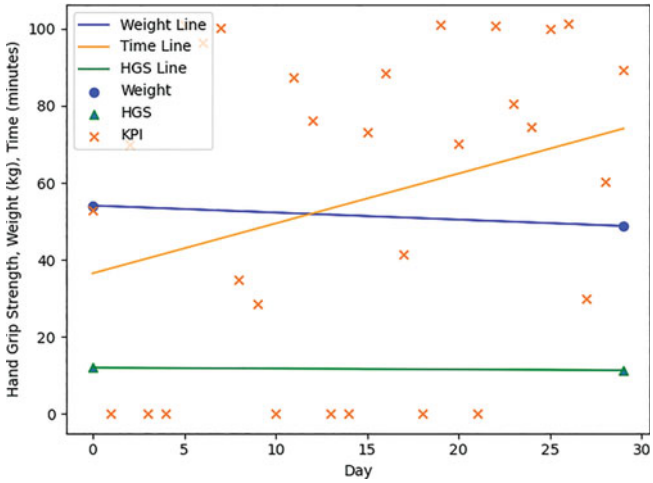


Figure 3.5 The second month of participant 4. Significant loss in body weight (54.1–48.8 kg), decrease in HGS (12.0–11.3 kg), increase in the KPI (≈39–76 min). HGS: Hand Grip Strength, KPI: Key Performance Indicator

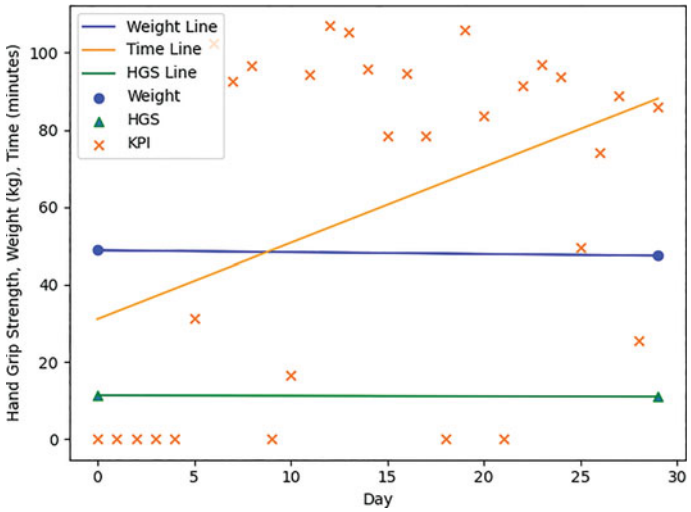


Figure 3.6 The third month of participant 4. Loss in body weight (48.8–47.4 kg), decrease in HGS (11.3– 11.0 kg), increase in the KPI (≈36–88 min). HGS: Hand Grip Strength, KPI: Key Performance Indicator

3.5 Discussion

The results showed a strong correlation between the HGS and the KPI as well as between the KPI and the body weight. Especially, in all month where a MCID occurs our method shows strong correlations. The last three months of participant 8 (Table 3.12) are all considered as MCID, even though there was no change in weight in the first two month of the three months interval. The loss in body weight was higher than 5% in the last month only and hence alarmingly. Participant 4 had a MCID in all observed months and even though only two power consumption sensors and one motion sensor were available for the analysis, strong correlation were found in each month. The diary revealed that the mental condition of participant 4 was not well and became worse in the considered months. After the last considered month the participant was admitted to the hospital and after discharge the participant moved to an assisted housing scheme. The loss in body weight might be a consequence of decreasing cognitive function and mental condition, concluding causalities from these correlations would be too early. Reasons are that neither combination of positive and negative correlations is consistent, nor

the magnitude is. In general, there are all possible increasing and decreasing combinations for the HGS, KPI, and body weight and also the sign of the correlation coefficient. Therefore, concluding causalities is not possible. A lot of factors may cause the change in one value. For example if the HGS is decreasing, and the KPI is decreasing as well, one reason could be the participant prepares simple meals, another reason could be that the participant eats pre-prepared meals or convenience food. Other possible explanations are the absence of ground truth label available for the iADL cooking and the heterogeneous sensor setup. The minimum set of sensors installed in a kitchen were one motion sensor, one door contact sensor in the fridge, and one power consumption sensor at the kettle. This may also resulted in the short average meal preparation time per day of 10.62 min. It is easy to construct a situation where a meal is prepared, but not two or more sensors are active. It is also easy to construct a situation where all sensors are active, but no meal is prepared. Therefore, the computed KPI is prone to measurement errors. Moreover, the body weight may have measurement inaccuracy, because it was not measured at the same time of the day and with various apparel. The diaries, the exclusion criteria *self supporter* was based on, were probably not completely precise, because the diaries were kept retrospectively and information were added based on participant's statements. Participants were probably embarrassed to admit being assisted or uncapable of supporting themselves. Furthermore, there might also be a possibility that there are external factors which were not kept in the diaries.

Overall, there are variety of factors which can influence the change in body weight and information about the nutritional value of the prepared meals (e.g., minerals, vitamins), which were not tracked in the study, would give better insights in the nutritional status of the participants.

3.6 Conclusions

The aim of this contribution was to show that the meal preparation time (KPI) is associated with body weight changes as an indicator for malnutrition. The results showed that the KPI measured by home automation and power sensors is associated with the change in HGS and the change in body weight measured by professional staff. Using this association one indicator for impending malnutrition may could be detected at an early stage. However, it is too early to conclude causalities from the correlation. For finding whether there are causalities or not, a study designed for that specific purpose is needed. The study should at least

survey the meal preparation time, the nutritional status, and the nutritional intake of the participants.

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Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Availability Statement: The data is not publicly available due to privacy concerns.

Conflicts of Interest: The authors declare no conflict of interest.

Supplementary Materials: The following are available online at https://www.mdpi.com/article/10.3390/nu13061955/s1 Table S1: 1STROBE Statement—Checklist of items that should be included in reports of cohortstudies.

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4 Using Sensor Graphs for Monitoring the Effect on the Performance of the OTAGO Exercise Program in Older Adults

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Abbreviations

GPS	Global Positioning System
IMU	Inertial measurement unit
PIR	Passive infrared
SD	Standard deviation
SPPB	Short physical performance battery
TUG	Timed Up and Go

4.1 Introduction

The frailty syndrome describes the condition in which the muscular performance and well-being of geriatric people continuously decline. This can result in a loss of independence and an increased risk of falls. Particularly, in the group

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of older adults, falls can have serious physical consequences and weaken the confidence in one's abilities, leading to a loss in independence. The exercise program named OTAGO aims to reduce the risk of falling with the help of an exercise program comprising 17 exercises tailored for the individual participant [1, 2]. It includes exercises in increasing difficulty to strengthen the lower-limb muscles, exercises to improve balance, and a training plan for regular walks. Risk assessment and performance monitoring can be conducted using well-established geriatric mobility assessments, which are performed under the supervision of healthcare professionals. Two common and validated mobility assessments are the Timed Up & Go (TUG) and the Short Physical Performance Battery (SPPB), each measuring the dimensions gait speed, lower-limb strength, and balance [3, 4].

Supervised assessments are often time-consuming and create a test situation in which the participants attempt to do their best (Hawthorne effect) [5]. However, this may lead to results that reflect capacity rather than the participant's performance [6]. The gap between capacity and performance of the mobility of older adults is a well-known fact in geriatrics research and studies show that performance is a better indicator of health condition than the capacity. The example gait speed shows that the self-selected gait speed is a good predictor for mortality rates and daily ambulatory activity in older adults [7–9]. Our objective is to measure the performance of older adults in daily life and compare the performance of the intervention group doing the OTAGO exercise program and the control group not doing the exercise program. We used passive infrared (PIR) motion sensors installed in the domestic environments and a weighted directed multi-sensor graph as an activity measure to achieve our objective. We used activity as surrogate variable for performance. PIR motion sensors were chosen because they are more privacy-preserving than, e.g., cameras. The capacity was measured by monthly standardized geriatric assessments SPPB and TUG. Our data was collected during the pilot observational "OTAGO" study with 20 (pre-)frail participants (aged: 84.75 y, SD: 5.19 y) over 10 months.

4.2 State of the Art

In this section, we give a brief overview of recent research. At first, we introduce the state of the art in geriatric medicine for assessing the mobility and risk for falls in older adults. We use these assessments as capacity measure in testing situations. Afterwards, we mention the computer science approaches to estimate mobility, gait parameters, and the falling risk.

The TUG test and the SPPB are two common validated mobility assessments in geriatric medicine [3, 4]. Both assessments assess the three dimensions gait speed, lower-limb strength, and balance. The procedure for the TUG test is as follows: the subject sits on a standardized chair, on the command "go", the subject stands up, walks three meters, then turns around, walks back to the chair, and sits down again. The supervisor measures the time in s using a stopwatch. The measurement starts when the supervisor gives the command and ends when the subject's buttock touches the chair again. Depending on the time, the subject's mobility and risk of falling can be categorized as

- $\leq$10 s: normal, no mobility impairment.
- 11–19 s: minor mobility impairment, not relevant in everyday life.
- 20–29 s: mobility impairment.
- >30 s: severe mobility impairment, need for intervention.

If an assistive device for walking is used, the usage must be recorded and the assistive device must be used in any retest. The dimension gait speed is assessed by the threemeter walk, the dimension balance by standing up from the chair and turning around, and lower-limb strength by standing up from the chair and sitting down again. While the TUG assesses the three dimensions implicitly, the SPPB has separate items for each dimension which are scored independently. The SPPB is comprised of the three items:

- Stance, semi-tandem stance, full-tandem stance (balance).
- Four-meter walk (gait speed).
- Five-time chair rise (lower-limb strength).

Each item can be scored from 0 points to 4 points, and a maximum score of 12 points can be achieved. The time needed for the four-meter walk and the five-time chair rise is measured in s by the supervisor using a stopwatch, and the balance is scored by the time the subject can stand in the different stances. The cut-off value for fit and frail people is $\leq$9 points and the minimal clinically important difference is 0.5 points [10, 11]. For medical purposes, the MCID is more meaningful than a statistical significance. Both assessments assess a limited number of parameters, e.g., jumping performance, walking backwards, lying down, and rolling are neither included in the SPPB nor in the TUG. Naturally, assessment results measure the form on the day. Moreover, it has been shown that energy and fatigue from, e.g., answering questionnaires could have an negative impact on the assessment results [12].

Researchers utilized various sensors to estimate gait parameters, mobility, and the risk of falling in older adults. There are mainly two different approaches. The first approach is to support the assessment using sensors and computer science technologies, and the second approach is to measure the parameters in daily life. The measurements in daily life are the closest to measure the performance instead of the capacity.

The movement during the TUG test can be found in several situations in daily life, e.g., standing up from a chair, sitting down on a chair, and walking a short distance are everyday activities. Therefore, inertial measurement units (IMU) were used to detect the TUG movements in everyday life and to estimate the time of the movements [13]. Moreover, an approach to estimate the scores of mobility assessment based on real-life IMU data has been made in [14]. The systems developed in [15, 16] are supposed to enable people to perform assessments in an unsupervised fashion.

As gait parameters and body movements can be measured directly when a sensor is attached to the body, it is not far to seek to utilize IMUs. Most of the research using wearable sensors are focused on gait parameters [17–19]. Studies using IMUs showed that the results are comparable to the gold-standard measurements [20–22]. In combination with deep neural networks, IMUs show promising fall detection and fall risk prediction results [23–26]. Even for older adults suffering from neurological disorders and multiple sclerosis, the results are convincing [27, 28]. The approaches described in [29–31] were able to measure gait parameters and walk characteristics using IMU and force sensor data. IMUs ambient sensors installed in domestic environments are in the focus of researchers. We only mention approaches that use privacy-conserving sensors because the sensors are installed in private areas. Privacy is of great concern, and cameras or light detection and ranging (LiDAR) sensors are too intrusive. Moreover, unobtrusive ambient sensors are well-accepted among older adults [32, 33]. Sensor networks installed in smart homes were used to estimate the self-selected gait speed in domestic environments [34]. Moreover, utilizing homogeneous networks of PIR motion sensors for the same purpose show precise estimation as well [35–38]. Another approach using radio signals estimated not only the gait speed but also the stride length as gait parameter [39].

The mentioned approaches primarily focus on gait parameters, and the technologyassisted mobility assessments place the participants in test situations again. However, our approach is more versatile and does not depend on special movements.

4.3 Materials and Methods

Study Design

The primary aim of the study was to analyze the "effect of the fall prevention intervention program OTAGO on older persons with frailty and prefrailty" and compare the conventional assessment of mobility and independence of older adults in (pre-)frail condition with sensor-based documentation. The study was conducted in 2014 and 2015 for a period of 10 months. A total number of 20 participants (17 female, 3 male), aged 84.75 y (SD: 5.19 y), were recruited at local nursing homes and among participants of previous studies. They were randomly assigned to the control and the intervention group. The cohort size was limited by human resources and funding. The inclusion criteria were as follows:

1. Age $\geq$ 75.
2. At least pre-frail by the definition of Fried (Frailty Index $\geq$ 2) [40].
3. Living alone within the city limits of Oldenburg.
4. Able to move inside the flat as needed.

The exclusion criteria were as follows:

1. Not able to move inside the flat as needed.
2. Keeping pets that move freely inside the flat.
3. Living with other people.
4. Severe visual impairment.
5. Contraindication against the OTAGO exercise program.
6. Unable to understand the purpose of the study and the study itself.

The average participation time was 36.5 weeks (no follow-up). Two participants passed away during the study. The participants in the intervention group were required to independently perform the 30 min OTAGO exercise program once a week and to walk for at least 30 min every day [1]. The exercise program includes five exercises to strengthen the lower-limb muscles, twelve exercises to improve balance, and a training plan for regular walks. The exercises could be performed in up to four levels of difficulty and were individually tailored to the participant. The execution and tailoring were controlled and carried out by licensed physiotherapists. In addition, in-home assessments were conducted once a month, during which the exercise selection and intensity were adjusted if necessary. The average time between two in-home assessments was 31.3 d (SD:

5.3 d). The in-home assessments included, but were not limited to, the mobility assessments Tinetti [41], SPPB, TUG, and the standardized questionnaires EQ–5D–5L [42], Mini Nutritional Assessment [43], and Instrumental Activities of Daily Living [44], amongst others. The questionnaires and assessments were selected to obtain a holistic view of the participants and their condition. The baseline was established by the first assessments at the beginning of the study; 10.35 assessments are available on average per participant. Tables 4.1 and 4.2 show a selection of characteristics of the study cohort and the used sub-cohort at baseline and the end of the study.

Table 4.1 The baseline/end characteristics of the cohort

n = 20/18 (m = 3/3, f = 17/15)	Age (y)	Frailty Index (Points)	SPPB (Points)	TUG (s)
Mean	84.75/85.44	1.90/2.00	5.95/6.61	17.87/16.12
SD	5.19/4.92	0.72/0.97	2.33/2.85	5.33/5.85
Range (min–max)	76.00–92.00/ 77.00–93.00	1.00–3.00/ 0.00–4.00	3.00–11.00/ 2.00–12.00	11.16–31.63/ 8.15–30.06

Table 4.2 The baseline/end characteristics of the used sub-cohort at the beginning of the study

n = 15/15 (m = 1/1, f = 14/14)	Age (y)	Frailty Index (Points)	SPPB (Points)	TUG (s)
Mean	84.60/85.20	1.80/1.73	6.53/7.07	16.81/14.87
SD	5.57/5.41	0.68/0.80	2.36/3.01	4.35/5.82
Range (min–max)	76.00–92.00/ 77.00–93.00	1.00–3.00/ 0.00–3.00	3.00–11.00/ 2.00–12.00	11.16–24.06/ 8.15–30.06

Data Acquisition

The sensor-based documentation was realized by installing an ambient multi-component sensor system in the domestic environments of the participants. In addition, two wearable sensors were given to them as well. The sensors for the ambient multi-component sensor network were chosen with privacy and acceptance among older adults in mind [32, 33]. The chosen sensors were PIR motion

sensors, door contact sensors, a four-key button, a vibration sensor, and power consumption sensors. The PIR motion sensors had a cool time of 8 s when motion could not be detected. The PIR motion sensors were installed so that they could cover the rooms of the flat and a person entering a room can be detected as soon as possible. Additional ones were placed so that a specific path in a room or between two rooms is covered, e.g., the way from the bedroom to the kitchen. One PIR sensor was placed in the bathroom right over the lavatory flush to detect if the resident used the toilet. Moreover, one redundant sensor was installed in every flat. The PIR motion sensors had a cool-down time of 8 s and measured motion in 8 s intervals. The door contact sensors were installed at the entrance doors to the flats and the fridge to detect when a door was opened. The button was placed at the front door and the participants were asked to push the button if somebody else was entering or leaving the flat. The door contact sensor and the button were combined to detect if another person was in the flat. The vibration sensor was placed between the slatted frame and the mattress of the resident's bed. Using the vibration sensor, sleep times information can be measured more accurately. The power-consumption sensors were attached to several appliances in the flat; these sensors were attached to the socket of the appliance and continuously measured the power consumption. All sensors transmitted their data by a wireless connection to a base station. The two wearable sensors were a Columbus V990 GPS sensor and an IMU of type Shimmer3r [45, 46]. The participants were asked to take the GPS sensor with them when leaving the flat. The IMU was given to the participants on every assessment day and collected one week after, and the participants were responsible for transferring the data and charging the IMU by placing it on a docking station. Only the data from the sensor network were used for this research, and no data from the wearable sensors. The sensor network differs from flat to flat because of the different topologies.

Sensor Graph

We modeled the flats as weighted directed multigraphs. Each vertex represents one motion sensor or one room. We used directed graphs because we considered the order of the sensor events, e.g., a transition from the living room to the kitchen is different to a transition from the kitchen to the living room. In this regard, we needed to use multigraphs, because there must be one edge for each transition. The weights were computed according to the order and number of motion sensor events, e.g., if there was a motion sensor event of the kitchen sensor followed by an event from the living room sensor, 1 was added to the weight of the edge

representing the transition from the kitchen to the living room. Hence, the primary diagonal of the adjacency matrix represents motion in one room. The motion in the same room was a loop in the sensor graph. The transitions are likely to be motion, and motion is essential for analyzing the effect of the OTAGO program. The primary diagonal had the largest values, because the participants were in one room and they were not changing rooms most of the time. To increase the importance of transitions, we used a weight matrix W with weight 1 on the primary diagonal and 1.5 elsewhere. In mathematics, a graph is defined as an ordered triple of vertices, edges, and an incident function $G = (V, E, \phi)$, where $V = \{s_0, \ldots, s_{n-1}\}$ is the set of vertices or sensors and n is the number of motion sensors in the flat, E is the set of edges, and $\phi : E \subseteq \{(s_i, s_j) \mid (s_i, s_j) \in V \times V, 0 \leq i, j < n\}$ is the incident function. Please note that we allowed loops in the graph by the definition of ϕ. We also introduced the edge weight update function $f_w(G, i, j) = G + T$, where $G \in R^{n \times n}$, $T \in \{0, 1\}^{n \times n}$ and 0 everywhere except for $t_{i,j} = 1$. The following Tables 4.3 and 4.4 show the baseline and a random day adjacency matrix of participant 6.

Table 4.3 The matrix is the baseline adjacency matrix of participant 6

	Bedroom	Bathroom	Kitchen	Living Room	Hallway
Bedroom	45.14	0	0.14	34.84	1.86
Bathroom	0	114.29	0	1	22.86
Kitchen	0	0	144.86	38.14	0.43
Living room	35.14	1.29	38.43	294.43	50.71
Hallway	1.71	19.57	0.14	51.86	70

The sensor graph for the flat in Figure 4.1 is shown in Figure 4.2. The edge weights were the baseline weights. Each vertex is connected to all other vertices with two edges in the sensor graph. The example shows that the most activity was found in the living room and the highest number of transitions was from the living room to the hallway. The graph also contains transitions that were impossible due to the topology of the flat, e.g., in the example flat, a direct transition from the bedroom to the bathroom was not possible. The impossible transitions have the weight 0.

Table 4.4 The adjacency matrix of a random day of participant 6

	Bedroom	Bathroom	Kitchen	Living Room	Hallway
Bedroom	45	0	0	34	1
Bathroom	0	98	0	1	25
Kitchen	0	0	196	39	1
Living room	32	1	39	230	68
Hallway	3	25	1	66	64



Figure 4.1 The flat of participant 6. The boxes indicate the position of the motion sensors, and the shades the recorded area

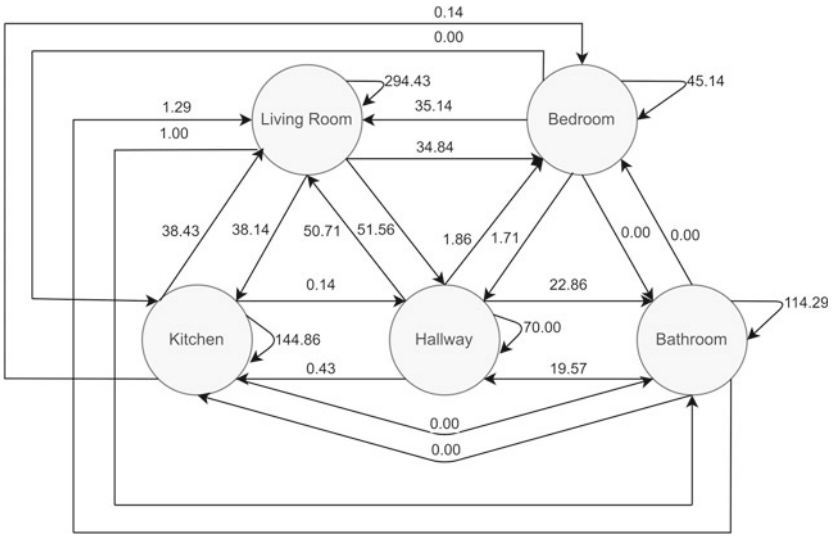


Figure 4.2 The sensor graph of the flat (Figure 4.1) of participant 6 with the baseline weights

Difference Function

Before computing the difference, both matrices, the baseline adjacency matrix B, and the daily adjacency matrix D, were elementwise multiplied with the weight matrix W. The difference is the standard subtraction defined over the ring of matrices:

$$\text{diff}(B, D) = \sum_{i,j=0}^{n-1} b_{i,j} - d_{i,j} \tag{4.1}$$

Finally, we converted the difference to percent p_d of the baseline:

$$pd = \frac{\text{diff}(B, D)}{\sum_{i,j=0}^{n-1} b_{i,j}} \tag{4.2}$$

Preprocessing

Before the algorithm was applied, the data must be preprocessed and filtered on participant scale and day scale. The participants were filtered by the following three exclusion criteria:

I. Passed away during the study.
II. Change in living situation.
III. Incidents compromising mobility.

Two participants were excluded according to criterion 1, they were deceased 9 months after study began. According to criterion 2, two participants were excluded because the first participant moved to a nursing home after 4.5 months and the second participant had the flat renovated after 7.5 months. According to criterion 3, one participant was excluded because the participant had a severe fall incident in the first month of the study and had to use a wheelchair after rehabilitation. Eventually, our analysis contained 15 participants; nine were from the intervention group and six from the control group. The days where the participant was absent were filtered, and 115 d were filtered, then overall, 4534 d (per participant: mean 302.27 d, SD 20.74 d) were left for our analysis. The reasons for being absent were vacation, short-term care, and hospitalization. One participant went on vacation and relatives were living in the flat during that time.

Statistical and Computational Methods

Before the analysis, we had to validate that our sub-cohort had no bias after applying the exclusion criteria. The dependent variables were the tests and questionnaire results. Thus, the dependent variables were all on metric scale and so the assumption for the test holds. We also report if there is a bias in the control group. We used the Wilcoxon signed-rank test to report if the difference in activity and the improvement in the mobility assessment scores were statistically significant. We carefully checked all assumptions for applying the test. Our computational method for capturing the tendency of the performance was the method of least squares. We used it to compute the best parameters a, b of a linear function.

4.4 Results

The statistical results showed that the disposition of the cohort and the sub-cohort were the same to a significance level of $\alpha = 0.01$ (Appendix A Table A1). The disposition of the intervention and control group were the same to a significance level of $\alpha = 0.01$ (Appendix A Table A2). The improvements in the SPPB and TUG score were not statistically significant for the control group and neither was there a difference in the intervention group.

The results for the geriatric assessments are shown in Figures 4.3 and 4.4. The fitted lines show that the capacity of both cohorts was increasing over time. The capacity of the intervention group was increasing faster than the capacity of the control group. The SPPB scores of the intervention group improved by 0.16 points, and the control group by 0.06 points. The lines of the relative TUG times were decreasing because the shorter the time, the better the mobility of the participant. The TUG time of the intervention group improved by −0.10 s, and the control group by −0.08 s. The differences were negative but meant an improvement in the test, and so in capacity. In Figure 4.5, the fitted lines of the activity difference to the baseline in % are shown. Both lines had a negative slope; the slope of the control group was smaller. The lines show that the activity in the flat of the control group was decreasing faster than in the flats of the intervention group. The difference between the intervention group was changing by −0.02% and of the control group by −29.35%. Both differences were negative, and that meant less activity in the flat. The parameters show that the slope of the fitted line of the control group is smaller by a factor of 10. The difference (H_0: the fitted activity of the control group has the same median as the fitted activity of the intervention group) is statistically significant to a significance level of $\alpha = 0.01$. The difference of the study duration (304 d) was analyzed and the test statistics of the Wilcoxon signed-rank test were $-4.76 < W = 46{,}360$, $p < 6.74 \times 10^{-52}$.

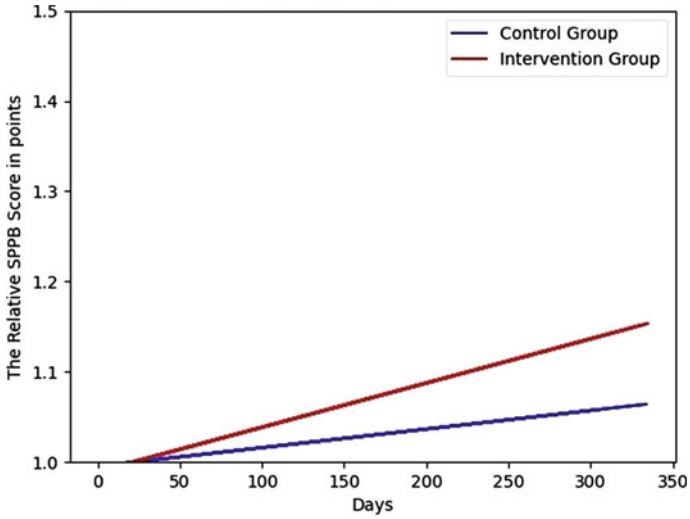


Figure 4.3 The fitted lines of the relative SPPB scores in points. The scale on the *y*-axis is the MCID

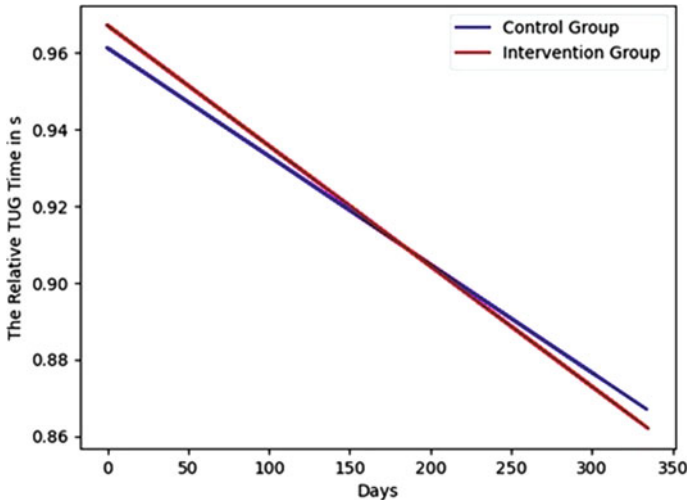


Figure 4.4 The fitted lines of the relative TUG time in s

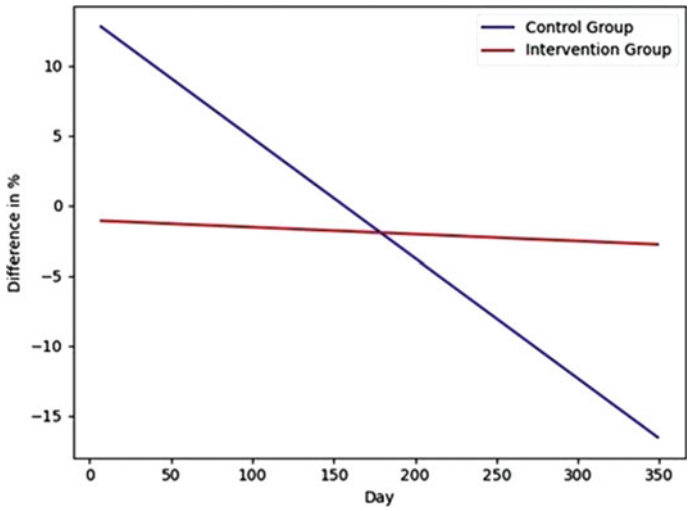


Figure 4.5 The fitted lines of the difference in % of the daily sensor graphs

4.5 Discussion

The analysis of the geriatric mobility assessments had expected results. The intervention group performing the OTAGO exercise program showed a greater improvement in capacity than the control group, and the analysis of the differences showed interesting results. The improvement in the control group may be caused by a learning effect. The differences of the sensor graphs were nearly stable for the intervention group, whereas the differences were decreasing for the control group. The analysis of the assessments showed that the magnitude of the difference of both groups is of factor 2, but the differences of the sensor graphs are a factor of 10. Considering the difference between capacity and performance places a different complexion on the difference analysis results. The results indicate that the OTAGO exercise program improves the capacity of older adults but not the performance. Furthermore, an improvement in capacity seems to not affect the performance in real life. On the one hand, an improvement in capacity is not visible because the person's actual performance is sufficient for daily life activities, e.g., a person would not run to the kitchen to get a cup, even though it would be faster. On the other hand, a decrease in capacity may not be visible until the level of capacity is decreasing below the level of performance.

Our results support the findings that the capacity does not reflect the performance and the Hawthorne effect [5, 6]. However, it is too early to conclude causalities from these results.

Several factors influence the activity and mobility in domestic environments. One factor may be the motivation to perform a specific task, e.g., going to answer the phone may be faster than going to the kitchen to return a used cup. During the study, some motion sensors were rearranged, disassembled, or dropped by the inhabitant of the flat. Moreover, the base station for recording the sensor data was disconnected or down. This led to faulty and corrupted data and complete loss of data. Hence, the groups are imbalanced after filtering. There are three fewer participants in the control group than in the intervention group. Another important limitation is the presence of other people in the flat. All participants were living alone in the flats, but they may have had visitors such as relatives, nurses, and workers. Filtering all the time where other people were present in the flat was not possible and the participants were not using the four-key button reliably. We tried to mitigate this limitation by fixing the weights of impossible transitions to 0, but if other people were in the same room, the activity may be overestimated. Another limitation is that the PIR sensors can only measure motion and not other aspects of mobility such as balance and lower-limb strength. The system could detect changes, and these changes could be used as an indicator for assessing the mobility using a standardized geriatrics assessment supervised by a licensed physiotherapist.

4.6 Conclusions

In this article, we introduced sensor graphs generated from PIR motion sensor networks in domestic environments that can quantify movement within, and transitions between, rooms and thus allow drawing inferences about the resident's performance. Our results indicated that sensor graphs generated from a PIR motion sensor network can be used to monitor the performance progress of older adults. In addition, we found an indication that the performance of older adults, who completed the OTAGO exercise program, declines slower than the performance of older people who did not complete the exercises, even though the capacity of both groups increased. The results also suggest that the OTAGO exercise program may improve the capacity of older adults but not their performance. However, it is too early to conclude causality from these results; hence, this will require further clinical studies and a sophisticated strategy for filtering the times when more than one person is in the flat.

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Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The data are not publicly available due to privacy concerns.

Conflicts of Interest: The authors declare no conflict of interest.

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5 Unsupervised Statistical Concept Drift Detection for Behaviour Abnormality Detection

Friedrich B, Sawabe T, Hein, A. Unsupervised statistical concept drift detection for behaviour abnormality detection. Applied Intelligence. 2022. https://doi.org/10.1007/s10489-022-03611-3

5.1 Introduction

Behaviour monitoring for healthcare purposes is an important task and a change in behaviour may indicate a medical condition. This includes physical diseases as well as cognitive diseases like dementia [1–4]. To get a holistic view of the behaviour, unobtrusive privacy preserving sensors can be installed in domestic environments. Human behaviour is very volatile and difficult to analyse, but machine learning approaches showed some promising results in this domain [5]. One disadvantage is that a vast amount of data is needed for training, thus in some cases, collecting large datasets is expensive or even impossible. Moreover, labeling the data is time consuming and labels can be unreliable.

Unsupervised machine learning algorithms are not relying on labels but need more training data. Our real–world dataset, collected during the 10–months observational OTAGO study, suffers from both limitations. Smart home sensors were installed in the domestic environments of 20 (pre–) frail older adults aged 85.75y (SD: 5.19y) and diaries were written for each participant. The dataset is both small and has unreliable labels, that is why we noticed that the most promising approach was to use unsupervised machine learning. Our goal was to detect

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changing or drifting behaviour and abnormal behaviour. Concept drift detection is one approach for reaching our goal. considering the limitations of our dataset the unsupervised concept drift detection seemed to be a reasonable approach. We extended the common approach by a novel method to deal with small datasets. We used the Central Limit Theorem (CLT) and the generalisation abilities of Variational Autoencoders (VAE) to get an accurate estimation of the underlying random distribution of our feature set. The feature set contains the Kullback–Leibler Divergence (D_{KL}) of daily activity probability maps and a 7d baseline activity probability map [6]. If a new day was normal the baseline was updated with the new day. Our new method can be applied to all features that are satisfying the CLT assumptions. Our aim was to detect abnormal behaviour as indication for medical condition in older adults. Before applying our new method to our real–world dataset we validated it in two steps on an artificial datasets. The first step was to validate whether the VAE can approximate the underlying distribution more accurately than common approach using the sample mean and the sample variance. In the second step we show that our approach can detect the four basic concept drift detection types. After our method passed both validation steps we applied it to our real–world dataset to detect abnormal behaviour of older adults. In the next section we are giving an overview of unsupervised concept drift detection methods and machine learning approaches for behaviour anomaly detection. In Section 5.3 we establish the theoretical foundations of our approach and we describe the data acquisition process, the preprocessing steps, feature engineering, and our validation process in detail. In the results Section 5.4 we present our results and we discuss them in the following section. In the last Section 5.6 we draw conclusions and provide an outlook for future work.

5.2 State of the Art

In the field of machine learning several approaches using different algorithms have been introduced. Common approaches with good results for abnormality detection and activity recognition in smart homes were using Long-Short–Term-Memory (LSTM) networks, which take the time domain into account [7–9]. However, these supervised approaches are dependent on labeled data. In contrast, the approaches developed in [10–12] were probability density based. A statistical

hypothesis test was used to determine if a new sample or set of samples was drawn from the same distribution. The authors in [13] combined a statistical significance test with the k–Nearest–Neighbour (kNN) algorithm. The kNN algorithm was used to partition the samples in the sample space. The significance test was used to check if a drift was present. A Student–Teacher approach for unsupervised concept drift detection was introduced in [14]. The student model was trained to mimic the loss of the teacher model. When new unlabeled samples arrived the discrepancy between the student and teacher models were used as surrogate signal for the concept drift detection. A common unsupervised concept drift detection method was used for the final detection step. With time series data in mind [15] developed an algorithm which was a composition of three different steps, dimensionality reduction, random partitioning, and anomaly detection. The anomalies were detected based on frequency statistics and the proportion of current anomalies and the average of previously occurred anomalies. The average is updated over time and leads to an adoption of the algorithm. The afore mentioned research was not focused on medical and healthcare applications. In healthcare, particularly for the task of abnormal behaviour detection, unsupervised concept drift detection had been used. Using motion sensor data, a graph of the flat, and a transition matrix of an unsupervised behaviour monitoring system was developed in [16]. The features were the probability of room–to–room transitions and detention time in a room. If a person made a transition or stayed in a room with a low probability an anomaly was detected. The probabilities were updated over time and able to adapt to the person's behaviour. The method introduced in [17] focused on the drift in performing activities of daily living. A similarity matrix between all time windows consisting of a sequence of raw sensor events were computed. The similarity was defined as the intersection of two histograms derived from the time windows. Afterwards the results were clustered using Markov Clustering. For detecting if a new time window is a drift they used the Silhouette Index in combination with a threshold. In [18] the aim was to assess the risk of abnormalities, that is why rule–mining was used to identify the causes of abnormalities. The resultant rules were used in combination with a Markov Logic network to detect the risk of abnormalities. Temporal information for abnormality detection were used in [19]. Based on sensor events and activities logs the probabilities of the temporal relation of activities were identified, e.g. the probability of one activity happens after a certain other activity. If a combination with a low probability occurred a drift was detected. The approach introduced in [20] was based on an unsupervised LSTM autoencoder. The encoder was trained on a baseline dataset containing motion sensor data, then the trained autoencoder was used to identify abnormalities by evaluating the deviation from the learned

regular model. The authors in [21] tried to detect abnormal days by comparing the activities performed on the day to the previous days. In the first step they used a machine learning approach to classify the activities based on smart home sensor data and in the second step they computed the boundary between normal and abnormal days by expanding the set of normal days. When the expansion stops the model was used for further classification. A more general review in [22] found that unsupervised deep learning methods outperform the (semi–) supervised methods in the task of abnormality detection in human behaviour.

We contribute an approach using a VAE as basis for a statistical test for concept drift and accurately estimating the parameters for a probability distribution for small datasets to the field of machine learning. We contribute to the healthcare domain by applying our method to unstructured smart home data of 20 (pre–)frail older adults for anomaly detection.

5.3 Materials and Methods

Data Acquisition

The data of 20 participants (17 female, 3 male) was collected during the observational OTAGO study conducted by the University of Oldenburg in 2014 and 2015 over a period of 10–months [23]. Due to dropouts the average participation time became 36.5 weeks. The average age of the participants was 84.75y (SD: 5.19y) and they were pre–frail or frail by the definition of Fried [24, 25]. The participants lived alone in their own flats, but there might be a possibility that they were assisted by a care giver. Every four weeks the staff from the University of Oldenburg visited the participants for supervising geriatric assessments, completing questionaries, and a diary. Holidays and time conflicts led to an average of 31.3d (SD: 5.3d) between two the visits. The following assessments for assessing the mobility and physical function were performed Tinetti [26], Short Physical Performance Battery (SPPB) [27], Timed Up & Go (TUG) [28], and Hand Grip Strength (HGS) [29]. For assessing the independence, quality of life, and nutritional status the questionaries Instrumental Activities of Daily Living (IADL) [30], EQ–5D–5 L [31], and Mini–Nutritional Assessment (MNA) [32] were completed. The EQ–5D–5 L and the MNA were only completed twice, at

the beginning and the end of the study. Tables 5.1 and 5.2 show selected characteristics of the cohort at the beginning of the study (T0) and the end of the study (T10). In addition to the regular visits and assessments, a variety of sensors were installed in the flats of the participants.

The sensor set contained home automation and power consumption sensors. Additionally, two wearable sensors, Shimmer3r (Inertial Measurement Unit) [33] and Columbus V990 (GPS), were given to the participants. A four key switch was installed next to the main entrance door and the participants were asked to turn it, when other people were entering or leaving the flat. The power consumption sensors were attached between the appliance and the power socket and measured all energy consumption of the appliance in W. The home automation sensors were comprised of passive infrared motion (PIR) sensors, contact sensors, and concussion sensors. The motion sensors were installed in the rooms to detect motion in a certain area and they had a scan–dead time of 8 s. Some sensors were installed to measure specific walk paths inside the flats, e.g. the way from the bedroom to the kitchen. One motion sensor was placed right over the lavatory flush to detect toilet use. The contact sensors were installed at all entrance doors of the flats and at the fridge. All home automation sensors had a wireless connection to a base station to transmit their data. The sensor setup was dependent on the topology and the wishes of the participants. A flat of a participant of the OTAGO study is shown in Fig. 5.1.

Table 5.1 The baseline characteristics of the cohort

n = 20 (m = 3, f=17)	Age(y)	Frailty Index(pts.)	SPPB(pts.)	TUG(s)
Mean	84.75	1.90	5.95	17.87
SD (±)	5.19	0.72	2.33	5.33
Range (min–max)	76.00–92.00	1.00–3.00	3.00–11.00	11.16–31.63

Table 5.2 The characteristics of the cohort at the end of the study

n = 18 (m = 3, f=15)	Age(y)	Frailty Index(pts.)	SPPB(pts.)	TUG(s)
Mean	85.44	2.00	6.61	16.12
SD (±)	4.92	0.97	2.85	5.85
Range (min–max)	77.00–93.00	0.00–4.00	2.00–12.00	8.15–30.06

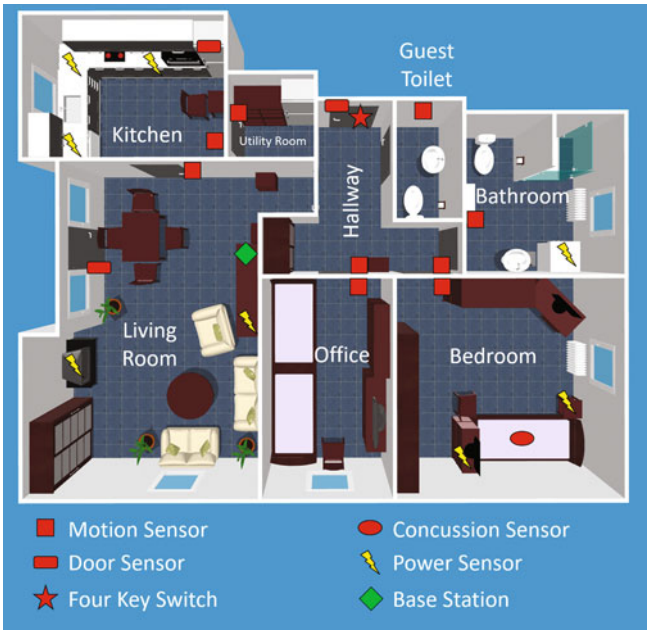


Figure 5.1 A flat of a participant of the OTAGO study

Data Preprocessing

Data preprocessing is an important step in machine learning application. That holds especially for medical data, where very often values are missing and samples are incomplete. Extensive preprocessing using imputation methods showed good results [34]. However, the concussion sensor was identified as unreliable and faulty in a way that imputation methods were not applicable. Therefore, we excluded it from the further analysis. One participant reported a fallen motion sensor and the sensor was fixed after. We did not exclude the data, because we were also interested in how the results were affected. The main preprocessing step was filtering and resampling the power sensor consumption data. The power consumption sensors were measuring standby energy consumption as well. Since the standby state did not contain any information about activity of a participant,

standby times were filtered. We assumed that the appliances were in standby most of the time. To find the standby power consumption of each appliance, we used a histogram to find the dominating W consumption and considered all values smaller or equal as standby power consumption. The motion sensors were sampling at 1/8 Hz and slower than the power consumption sensors. So, we resampled the filtered power consumption sensor data to 1/8 Hz.

Activity Probability Maps

An activity probability map contains the probability of having a sensor event for each hour of the day. The probability an event of sensor i occurring in hour h was computed by $p(s_{h,i}) = \#s_{h,i}/\#s_i$, where s_i was the sensor of interest, $\#s_{h,i}$ the number of events in hour h of sensor i, and $\#s_i$ the total number of sensor events of sensor i for the current day. The maps were created for each room of the flat separately and using a subset of all sensors. Figure 5.2 shows an activity probability map of a kitchen and Fig. 5.3 of a living room. We established average activity probability maps for each room as baseline for our analysis. The baseline map was the average probability of the maps of 7 days to capture each weekday once. We carefully checked the diaries and none of the participants had any medical condition or health related incidents in the first 7 days.

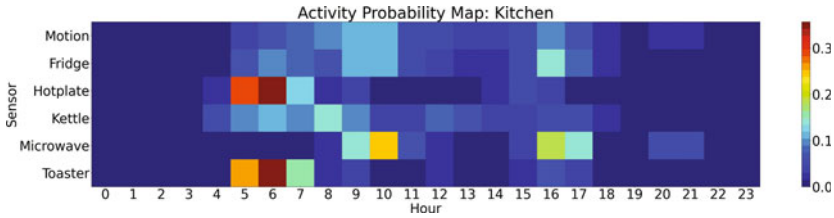


Figure 5.2 One example of an activity probability map of a kitchen. Hours 5 and 6 show high probabilities for using the toaster, the hot plate and the kettle

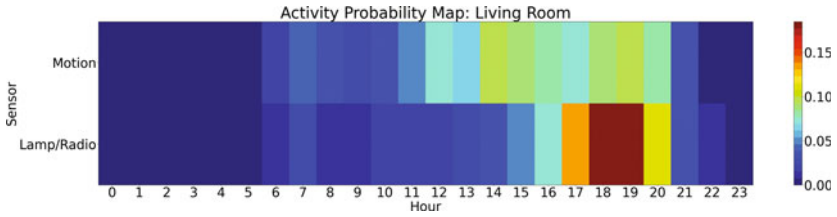


Figure 5.3 One example of an activity probability map of a living room. The power consumption sensor was attached to a multi socket where a lamp and a radio were connected to

Feature Engineering

The feature that was used for training the VAE was the statistical distance measure D_{KL}. The D_{KL} is defined as follows

$$D_{KL}(P||Q) = \sum_{x \in X} P(x) \log\left(\frac{P(x)}{Q(x)}\right) \tag{5.1}$$

where X is the probability space, and P and Q probability distributions defined over X. Here P and Q were activity probability maps. In case the quotient becomes 0, using

$$\lim_{x \to 0^+} x \log(x) = 0 \tag{5.2}$$

results in a contribution of 0 to the sum, if $P(x) = 0$. We computed the D_{KL} for each sensor of the activity probability map and the baseline. The final distance was the average over all D_{KL} of all rooms. The final feature set was scaled to the interval [0,1]. One benefit of the D_{KL} is its robustness against small probability changes. A small change in the probability map leads to a small change of the D_{KL}.

Unsupervised Statistical Concept Drift Detection

Concept drift detection is a method for detecting drifts and abnormalities in data streams and time series data. The basic idea is to compare new data to old data using a similarity measure and train a classifier. Concept drift detection can be

used according to the supervised learning or the unsupervised learning paradigm. In supervised concept drift detection it is assumed that the label is available right after the new data arrived and then based on the label the classifier is updated. However, in case the label is not available, unsupervised learning must be used. In the unsupervised way the decision whether a drift was detected is usually based on a user defined threshold. If the similarity measure exceeds the threshold a drift detection alert is raised. We adapted the general process to our approach and call it *unsupervised statistical concept drift detection.* Figure 5.4 illustrates the process of concept drift detection.

In the beginning we established a baseline activity probability map as described in Section 3.3 and computed the corresponding features (Section 3.4) to generate the baseline feature set. Then 100,000 samples for training and 50,000 samples for validation are drawn from a normal distribution $N(\mu,\sigma^2)$ parameterised by the sample mean and sample variance. The VAE is trained and validated on the sampled data. The prior, the latent space and posterior were normal distributions, because the CLT establishes that the limiting distribution of a set of random samples is a normal distribution. In the next step, *Concept Drift Detection*, the hypothesis H_0: *The new sample is not a drift* is tested. The cumulative distribution function of the posterior normal distribution given the new sample s is evaluated and if $P(X \leq s) \leq \sigma$ the hypothesis is accepted and rejected otherwise. If H_0 is accepted, the new sample s is added to the baseline feature set and hence the baseline feature set is updated and the process starts over for the next sample. If the hypothesis is rejected an alarm for a detected drift is raised.

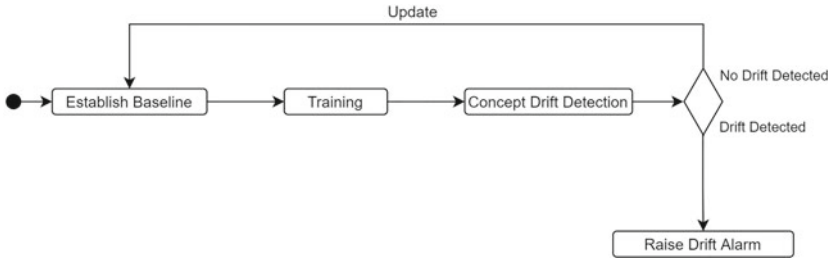


Figure 5.4 The illustration of our unsupervised statistical concept drift detection approach

Central Limit Theorem

The CLT establishes that the limiting random distribution of a sequence of identical independent distributed (i.i.d.) random variables is a standard normal distribution. The samples must be drawn from a distribution with finite mean and finite variance. The mathematical formulation of the Lindeberg–Levy CLT is stated as follows: Let $\{X_1,\ldots,X_n\}$ be a sequence of i.i.d. random variables with $\mathbb{E}[X_i, \ldots, X_n]$ and $\mathbf{Var}[X_i] = \sigma^2 < \infty, i = 1, \ldots, n$ $\lim_{n\to\infty} \sqrt{n}(\overline{X}_n - \mu) = \mathcal{N}(0, \sigma^2)$ In general, n must be a sufficient large number, but the CLT itself does not make a statement about the size of n. Since our sample size n is limited we used an VAE to approximate the limiting distribution stated by the CLT with a small n.

Variational Autoencoder

An autoencoder is a special type of artificial neural network that learns to efficiently encode data in a latent space. The network is comprised of an encoder and a decoder. The encoder is mapping the input data to a latent space and the decoder is reconstructing the data. The autoencoder is trained in an unsupervised manner by comparing the input data and the corresponding decoded data using a similarity measure. The autoencoder is learning a latent representation of the data. A VAE is a special type of autoencoder and learns the distribution of input data and hence encodes the data as distribution over the latent space. For achieving that the latent space is regularised during the training process. The training process has four steps. The first step is to encode the input as a distribution over the latent space. The second step is to draw a sample z from the latent distribution. In the third step the sample is decoded and the reconstruction error is computed. In the last step, the network is updated by backpropagating the reconstruction error. The latent space must be reparameterised to enable the gradient updating the encoder [35]. Figure 5.5 illustrates the used VAE architecture.

The encoder learns the latent distribution $q_\varphi(z \mid x)$ of the posterior distribution $p_\theta(z \mid x)$, where it is assumed that the data is generated by a distribution $p_\theta(x \mid z)$. Here, φ and θ were the parameters of the encoder and the decoder respectively. In our case the prior distribution was $N(\mu_e,\sigma_e)$, where μ_e and σ_e were the sample mean and sample variance of our data. The resultant distributions are

$$q_\phi(z|x) = N(\mu_{enc}(x), \sigma^2_{enc}(x)) \tag{5.3}$$

and

$$p_\theta(x|z) = N(\mu_{dec}(z), \sigma^2_{dec}(z)) \tag{5.4}$$

where $\mu_{enc}(x)$ and $\sigma_{enc}{}^2(x)$ are encoder outputs and $\mu_{dec}(z)$ and $\sigma_{dec}{}^2(z)$ the decoder outputs. For evaluating H_0 the cumulative distribution function of $N(\mu_{dec}(z),\sigma_{dec}{}^2(z))$ was evaluated for x. We trained the VAE using the Evidence Lower Bound Objective (ELBO).

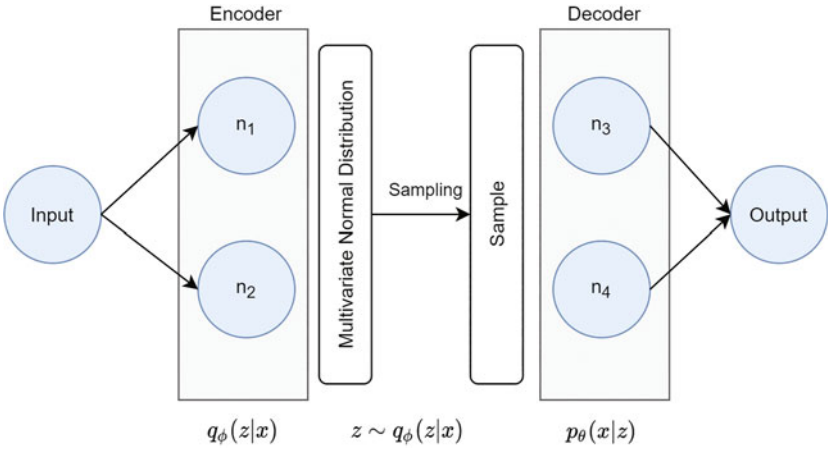


Figure 5.5 The illustration of our VAE. The used activation function was the sigmoid function and the used optimiser was ADAM with an initial learning rate of 1×10^{-4} and an exponential decay after the 5th epoch. The training was automatically stopped, if no improvement of the validation loss larger than 0.001 was found within 5 epochs

Method Validation

We validated our methodology in two steps. The first step was to validate whether the VAE is able to achieve a better approximation of the mean and the variance of the unknown underlying normal distribution than the sample mean and the sample variance. In the second step we validated whether our approach can capture the four common concept drift types. In practical applications the random distribution of the measurements is mostly unknown and must be approximated. The CLT shows that the limiting distribution Z for random samples $X_1,\ldots,X_n$ with mean μ and finite σ^2 is the standard normal distribution. The more measurements are

available, the better the approximation. We used a standard normal distribution N(0,1) as ground truth and drew two samples. Then we computed the difference between the approximated means and variances and the ground truth 1,000 times. We took the sample mean and the sample variance of all distances to judge the approximation quality of both approaches. To compute the mean and the variance of the VAE approximation we evaluated the VAE at a randomly sampled point from the approximated distribution $N(\mu,\sigma^2)$. To validate whether our method is able to detect concept drifts we created artificial datasets.

We created one dataset for the four typical concept drift types, sudden, gradual, incremental, and recurring described in [36]. Figure 5.6 illustrates the four different drift types.

Each set contained 300 samples on the domain [0,1] where 0 denotes no drift and a value larger than 0 a drift. The sudden concept drift data set contained 150 normal samples and 150 drifted samples. The drift occurred in the middle of the series after 150 samples. The gradual drift set contained 165 normal samples and 135 drifted samples. After 10 samples n drifted samples were inserted; n

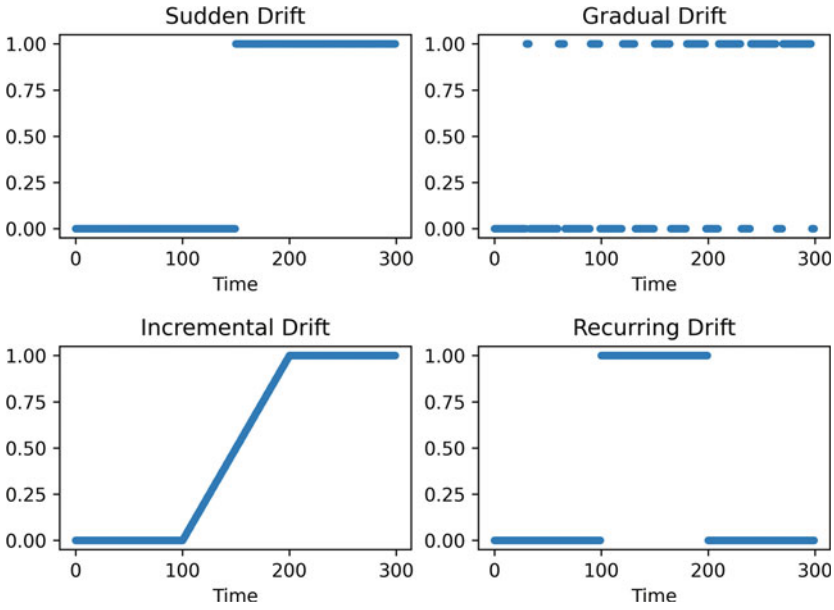


Figure 5.6 The four typical drift types we used for validating our method

counts from 1 to 9. The incremental drift set contained 100 normal and 200 drifted samples. The drift started after sample 100 and was linear incremented until sample 200. The recurring set contained 200 normal samples and 100 drifted samples. The samples between sample 100 and 200 were drifted samples.

5.4 Results

Underlying Distribution Approximation

Table 5.3 shows the results of the experiments for validating the VAE approximation of the underlying random distribution.

The average difference of 1,000 runs using two samples was 0.91 and the variance was 0.41, and the average difference using the VAE approximation was 0.73 and the variance 0.33. The results show that the VAE approximates the unknown underlying distribution more accurate than the sample mean and the sample variance.

Table 5.3 The results of the distribution approximation. The mean and variance rows show the mean and variance of the difference between the baseline {0,1} and the approximations

	Sampled	Approximation
Mean	0.91	0.73
Variance	0.41	0.33

Artificial dataset

The results of our second validation step are shown in Table 5.4.

The best scores were 1.0 on the sudden drift set and the recurring drift set. The worst scores of 0.91, 0.89, and 0.83 of the F_1-Score, accuracy, and recall were on the incremental drift set. The results on the gradual drift set were a F_1-Score of 0.98, an accuracy of 0.99, and a recall of 0.97. All the scores are referring to the artificial dataset. The lack of labels prevented calculating scores on our real–world dataset.

Table 5.4 The results of the concept drift validation

	Sudden	Gradual	Incremental	Recurring
F_1–Score	1.0	0.98	0.91	1.0
Accuracy	1.0	0.99	0.89	1.0
Precision	1.0	1.0	1.0	1.0
Recall	1.0	0.97	0.83	1.0

Real–world Dataset

Figures 5.7 and 5.8 show the distributions parameterised by the sample mean and sample variance and parameterised by the parameters estimated by the VAE. The red histogram is the baseline distribution of 7d and the blue histogram is the final distribution after processing all data.

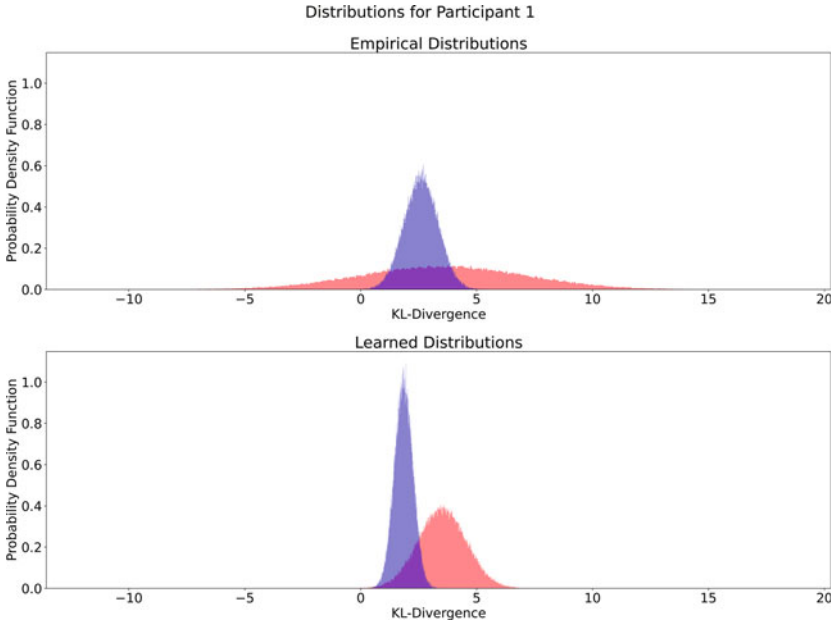


Figure 5.7 The drift of the distributions of participant 1. After beginning the chemotherapy no normal day is found using 1σ for testing the H_0. The left histograms show the distributions at the beginning and the right at the end of the study

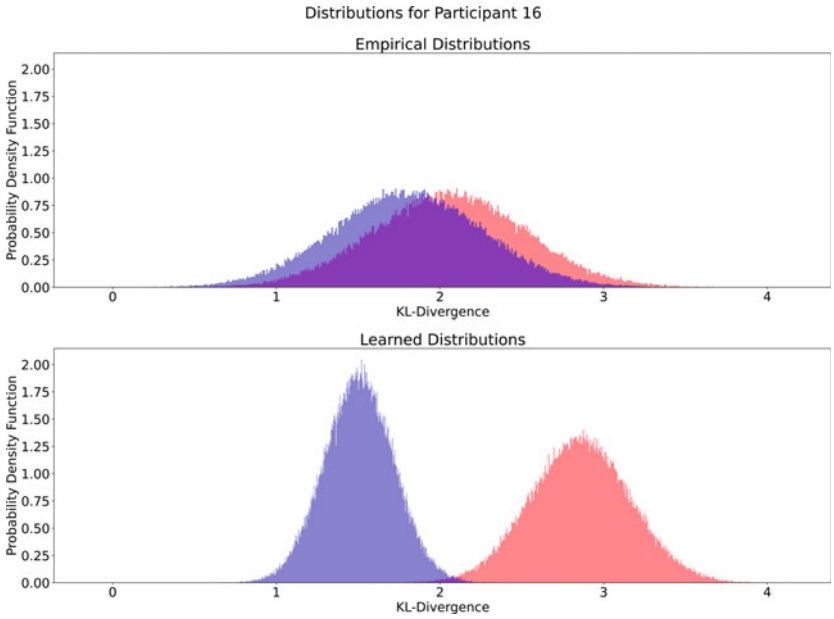


Figure 5.8 The drift of the distributions of participant 16. After beginning the chemotherapy no normal day is found using 1σ for testing the H_0. The left histograms show the distributions at the beginning and the right at the end of the study

The sample mean and the sample variance at baseline for participant 1 were 3.56 and 3.59. The mean and variance estimated by the VAE were 3.52 and 1.04. At the end the sample mean and the sample variance were 2.60 and 0.73. The final estimated mean and variance were 1.85 and 0.40.

After the first 66 days of the study participant 16 started a chemotherapy and afterwards no normal days were found anymore. The sample mean and variance at the beginning of the study were 2.06 and 0.47. The mean and variance estimated by the VAE were 2.86 and 0.31. The sample mean and variance at the end of the study were 1.81 and 0.46. The VAE estimation of the mean and variance at the end were 1.51 and 0.21. Table 5.5 shows the number of normal and abnormal days for different σ for each participant.

Table 5.5 The results of our statistical concept drift detection algorithm on the real–world dataset for different σ. normal days / abnormal days

ID	1σ	1.5σ	2σ
1	274/32	299/7	304/2
2	171/130	272/29	294/7
3	241/94	316/19	327/8
4	265/39	292/12	300/4
5	189/141	311/19	327/3
6	212/90	271/31	297/5
7	248/79	318/9	323/4
8	95/56	139/12	148/3
9	233/30	257/6	260/3
10	293/15	304/4	307/1
11	214/101	282/33	309/6
12	275/81	346/10	356/0
13	256/54	293/17	302/8
14	177/31	197/11	205/3
15	187/107	279/15	293/1
16	110/42	145/7	149/3
17	284/15	295/4	296/3
18	272/61	319/14	330/3
19	294/43	300/37	332/5
20	221/70	278/13	286/5

5.5 Discussion

The results on the artificial concept drift datasets and the VAE approximation were easy to judge, because ground truth values were available. Our real–world dataset did not have reliable ground truth values. First of all, we could see that the algorithm behaves as expected, when adjusting the factor of σ. The larger the factor of σ, the more a value can deviate from the mean without being considered as abnormal. The results showed as well that our method is able to adapt to the volatile behaviour of humans. A good indicator for the suitability of our method was participant 16. The participant got cancer during the study and had to undergo chemotherapy. The participant deceased before the study finished. Chemotherapy has severe side effects and is stressful for body and mind. Choosing 1σ our method did not detect any normal day after the first chemotherapy treatment. Our

approach was able to capture the change much better than the common statistical approach, because the shift of the mean was much larger. Another interesting case was participant 11. After a fall incident the participant had an injured ankle and wore an orthesis. The first 11 days were abnormal according to our algorithm, but then after the 11th day normal days were found again. Perhaps the participant adjusted to wearing the orthesis and the behaviour normalised again. Other participants were hospitalised during the study and those days were abnormal as well. Participant 3 lend the flat to relatives while on vacation, so all 5 days were recognised as abnormal. With 1σ our method considers all days where the participants had a medical condition as abnormal, but days where visitors came or the participants went on trips were considered as abnormal as well. One participant reported that a motion sensor fell off and all days until it was fixed by staff of the university were considered as abnormal. We found that all days participant 15 had an inflammation of the bladder and participant 16 had gastroenteritis were considered as abnormal. Even though the diaries were an unreliable ground truth source, we considered severe medical conditions like falls, bone fractures, hospitalisation, gastroenteritis, inflammation of the bladder, and chemotherapy as reliable. Other information like going out for a walk, having visitors, and playing games were not considered as reliable information. The algorithm considered days as abnormal where no reliable information is available. That is an indicator that our method has false positives. However, in the medical context a false positive is preferable over a false negative. Our system keeps a human in the loop and in case of an alarm the reason can be found. The combination our unsupervised concept drift detection method with a high precision and an medical expert with a high recall may lead to a high performance. There were the factor of σ and the days of the baseline to choose. The baseline of 7 days were chosen based on the assumption that routines on day scale are dependent on the day of the week. That may not be necessarily true. In clinical application the initial baseline would be 2 days because at least 2 days are needed to calculate the variance. Since the label would be available only normal days can be considered for the baseline. The factor of σ would be adapted in online learning fashion. The parameter controls the sensitivity of the model. In case of many false alarms the factor would be increased.

5.6 Conclusions

We introduced our novel unsupervised statistical concept drift detection method, and the results on the artificial dataset showed that our method was able to detect concept drifts using activity probability maps derived from home automation data and the D_{KL}. Moreover, we showed that our new approach using VAE for estimating the probability distribution is more accurate than the sample mean and sample variance. We evaluated our method on a real– world dataset of 20 (pre–) frail older adults and showed that our method was successful in adapting to a change in behaviour. Moreover, we showed that our method also detects abnormal behaviour in case of medical condition. Except for the medical conditions no reliable ground truth labels were available and so there may be false positives. Our approach was designed with humanised and personalised care in mind and keeps a human in the loop. In this context our approach is supposed to have a high precision and a high recall. The next is to test our approach on a similar dataset with reliable ground truth values. Considering the medical perspective a clinical study designed to evaluate our approach would be the most sophisticated way of testing our method.

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Data Availability Availability of Data and Materials The data is not publicly available due to privacy concerns.

Declarations

Conflict of Interests The authors declare no conflict of interest.

Informed Consent Informed consent was obtained from all subjects involved in the study.

Institutional Review Board The OTAGO study was conducted according to the guidelines of the Declaration of Helsinki, and approved by the Institutional Review Board of Carl von Ossietzky University (protocol code: Drs.27/2014, date of approval: 30.04.2014).

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6 A System for Monitoring the Functional Status of Older Adults in Daily Life

Friedrich B, Elgert L, Eckhoff D, Bauer JM, Hein A. A System for Monitoring the Functional Status of Older Adults in Daily Life, Scientific Reports. 2023. https://doi.org/10.1038/s41598-023-39483-x

6.1 Introduction

Older adults are prone to functional decline, this leads to an increased need of care and eventually, they might need to move to a nursing home. Moving to a nursing home has a negative effect on the three basic psychological needs competence, relatedness, and autonomy [1]. This also puts a financial burden on the older adult, relatives, and the health care system. In addition, there is a lack of medical professionals which affects the quality of care [2]. This issue becomes increasingly serious, as many nations are undergoing a demographic c hange [3]. One solution to face these challenges might be to enable older adults to live longer independent lives in their own domestic environments or in other words enable them to age in place.

For that purpose, we introduce a new Older Adults Monitoring System (OdAMS) that monitors the functional status of older adults in daily life and enables medical professionals to create individualised rehabilitation programmes that have a better outcome than generic programmes [4]. This helps older adults to maintain their functional abilities and live an independent life. OdAMS can also be used by medical professionals to monitor the effect of rehabilitation programmes. One of the benefits is the measurement in daily life without creating a testing situation. Real life mobility measurements tend to be better indicators and predictors than test scores, e.g. real life gait speed is a good predictor for the

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duration of hospitalisation and m ortality [5–8]. Many factors affect the health and functional status of older adults and that makes each individual an unique case [9, 10]. This fact is wellknown in geriatrics medicine and was found by analysing the sensor data from older adults collected in their domestic environments in daily life [11]. OdAMS was specifically tailored to the individual by being adapted using data from the individual only. The data basis was formed by inertial measurement units (IMU), home automation, and power consumption sensor data. Since the data was collected in one of the most private spaces of older adults, unobtrusive privacy preserving sensors had to be used. Moreover, this type of sensor is accepted by older adults amongst different cultures, especially for the purpose of keeping their a utonomy [9–14]. Three components linked the sensor data to the ICF categories: Activities and Participation (items: d450, d460), Body Functions (item: b530), and Environmental Factors (items: e115, e245) [15]. The ICF items were taken from the ICF Core Set: Geriatrics Comprehensive [16].

6.2 State of the Art

Using smart-home and wearable sensors in the medical context of older adults has come into focus of researchers. Much effort was given to conduct studies to get insights on the impact of such system and their benefit for measuring health i ndicators [17–21]. One main goal was activity and behaviour monitoring for the purpose of drawing conclusions on the health status [22–24]. While activity monitoring is unspecific, other researchers investigated the detection of serious specific incidents like fall detection [25–27]. A few attempts to measure the functional decline of older adults in daily life have been made. All of them are focusing on selected specific aspects [19, 24, 28–30]. However, focusing on one aspect does not give a holistic view on the older adult and neglects the complexity of the interaction between different aspects like physical functional decline and mental functional decline. Moreover, previous research does not link their outcomes and model outputs to the ICF and that may create a barrier for medical professionals and physician–researchers to use the results effectively. OdAMS addresses both disadvantages of the current state of the art. A comprehensive state of the art analysis for each sub-component can be found in the related articles [31–34].

6.3 Methods

The foundation of OdAMS is the data collected during the observational randomised pilot intervention study OTAGO [35]. The system was comprised of three components and used the ICF as output. In the this section we describe the study design, the system, and all of its parts. The data of the participants and the source code of the system can be found in the supplementary materials.

The OTAGO study was conducted according to the guidelines of the Declaration of Helsinki, and approved by the Institutional Review Board of Carl von Ossietzky University (protocol code: Drs.27/2014, date of approval: 30.04.2014).

Study design. The aim of the study was to investigate the "Effect of the fall prevention intervention program OTAGO on older persons with frailty and prefrailty—an analysis with conventional geriatric and technology–based assessments". The observational randomised pilot intervention study was conducted from 07/2014 to 06/2015. Data of 20 participants (17 female, 3 male) was collected during 36.5 weeks on average. Two participants passed away during the study. Considering the pilot study character and technology focused approach, the number of participants was sufficient. The inclusion criteria for the study were

1. Age ≥ 75
2. At least pre–frail by the definition of Fried [36]
3. Living alone within the city limits of Oldenburg
4. Able to move inside the flat as needed
5. And the exclusion criteria were

1. Not able to move inside the flat as needed
2. Keeping pets that move freely inside the flat
3. Living with other people
4. Strong visual impairment
5. Contraindication against the OTAGO exercise programme
6. Unable to understand the purpose of the study and the study itself

The participants were randomly assigned to the control group and intervention group. During the study every 4 weeks (avg. 31.3 days, SD: 5.3 days) a battery of geriatrics assessments was executed and physical parameters were measured in the homes of the older adults by a licensed physiotherapist. The assessments used for this research are Hand Grip Strength (HGS), Frailty Index (FI) [36], Short Physical Performance Battery (SPPB) [37], Timed Up & Go (TUG) [38],

instrumental Activities of Daily Living (iADL) [39], and the physical parameter body weight. Tables 6.1 and 6.2 are showing the characteristics of the study cohort at baseline (T0) and the end of the study (T10) (edited [31]).

The sensor system, which was installed in the homes of the participants, was comprised of home automation, power consumption, and two wearable sensors—one IMU and one GPS receiver. The power consumption sensors were installed between appliances and outlets, measuring the electricity consumption of the connected device. In terms of home automation sensors, passive infrared (PIR) motion sensors and door contact sensors were installed in each apartment. The door contact sensors were installed at the front door, windows, fridge, and the backdoor, if applicable. Additionally, a concussion sensor was placed in-between the slatted frame and the mattress. An extra four-key switch was placed near the front door and the participants were asked to turn the switch when another person enters or leaves the flat. All sensors send transmitted their data to a base station using a wireless connection. An example of a flat is shown in Fig. 6.1. Since each participant was free to choose the sensors, some sensor setups were different.

Table 6.1 Characteristics of the study cohort at baseline (T0) (edited [31])

n = 20, m = 3, f = 17	Age (years)	Body weight (kg) m/f	HGS (kg) m/f	FI (pts.)	SPPB (pts.)	TUG (s)	iADL (pts.)
Mean	84.8	66.2/69.2	17.1/14.1	1.9	6.0	17.9	7.3
SD (±)	5.2	3.6/17.3	5.1/7.0	0.7	2.3	5.3	1.4
Range (min–max)	76.0–92.0	61.9/ 43.8–70.8/ 115.9	11.7/ 3.7–24.0/ 33.0	1.0–3.0	3.0–11.0	11.2–31.6	3.0–8.0

Table 6.2 Characteristics of the study cohort at the end (T10) (edited [31])

n = 28, m = 3, f = 15	Age (years)	Body weight (kg) m/f	HGS (kg) m/f	FI (pts.)	SPPB (pts.)	TUG (s)	iADL (pts.)
Mean	84.5	66.1/71.0	16.8/12.2	2.0	6.6	16.4	6.1
SD (±)	4.9	5.4/19.1	4.2/3.7	1.0	2.9	6.0	2.3
Range (min–max)	77.0–93.0	59.1/ 42.7–72.1/ 123.7	13.3/ 5.0–22.7/ 18.0	0.0–4.0	2.0–12.0	8.5–30.1	1.0–8.0

International classification of functioning, disability and health. The ICF was developed by the WHO in 1980 and received approval from all member states in 2001. The main purpose was to establish a common standard for describing health and health-related states from individual scale to population scale. The benefit for patients is that not only the diagnoses are covered but also the physical, mental, and social aspects of the health condition [15]. Figure 6.2 illustrates the ICF's basic model of disability [15]. The holistic view on the patient's condition enables medical professionals to plan rehabilitation programmes considering more aspects than the diagnosis only. The ICF is comprised of too many items to be applied directly. Therefore, the ICF Core Sets were introduced as reference framework [16].

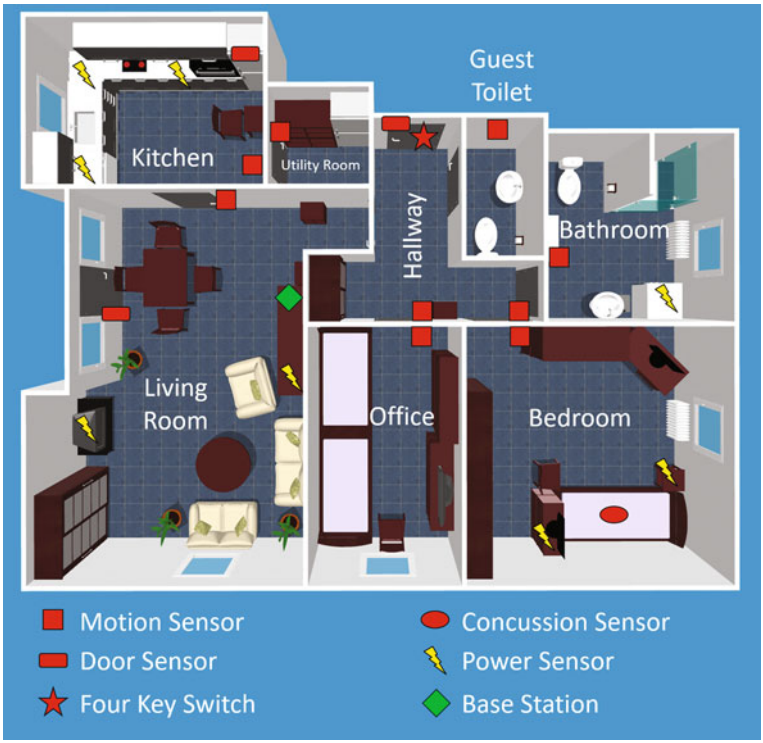


Figure 6.1 An illustration of one of the flats, which was equipped with the sensor system

System. The final system is illustrated in Fig. 6.3. Each component was linking data to one ICF category and one or two ICF items. The output of the first component monitored the assessment scores of standardised geriatrics assessments and the changes using IMU data. The second component monitored the change in body weight using home automation and power consumption sensors. The third components also used home automation and power consumption sensors but monitored the utilisation of electric appliances and the activity. The late fusion in the metric was done by logical or conjunction of the outputs. A change in one of the categories triggered an alarm and a medical professional could track the reason down to the ICF item and take the necessary measures. The metric and the components are described in the following subsections.

Metric. All three categories were considered as equally important and the metric was defined accordingly. The metric's input was the outputs C_1 , C_2 , C_3 of the three components and the outputs were concatenated by logical or ($\vee$). If one of the inputs was 1, i.e. one component was indicating a change an alarm was raised. No change was indicated by 0 by the components. Equation (6.1) shows the metric as Boolean algebraic expression.

$$C_1 \vee C_2 \vee C_3 \tag{6.1}$$

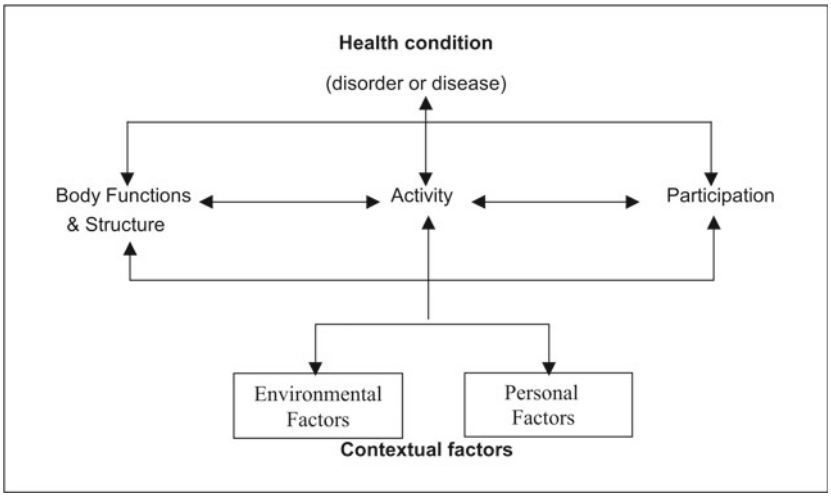


Figure 6.2 The disease model used as foundation of the ICF[15]

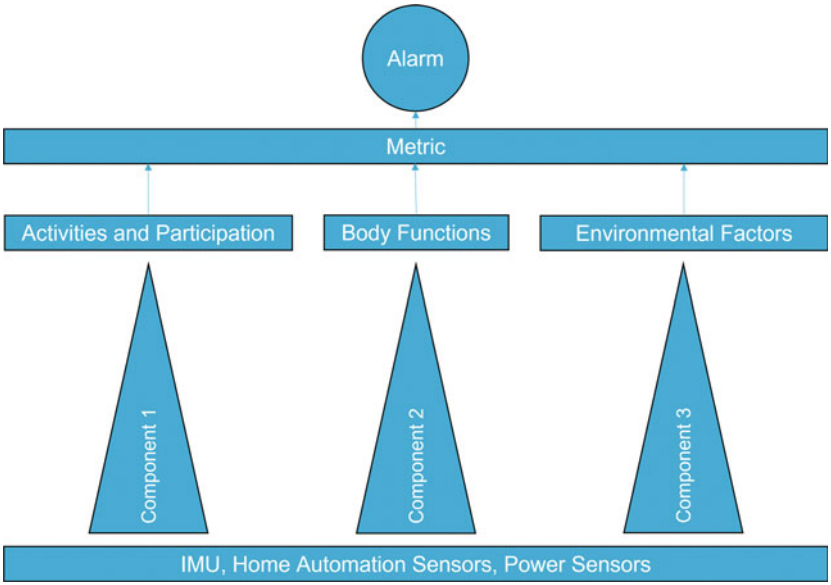


Figure 6.3 The final system comprised of three components

Component 1: activities and participation. The first component was comprised of two deep neural networks. The networks took the magnitude, i.e. Euclidean Norm of all three axis, of the IMU sensor's low-range accelerometer, wide-range accelerometer, gyroscope, and magnetometer as input and predicted the SPPB and the TUG scores, not the underlying raw values (e.g. seconds in case of the TUG), the older adult would achieve. The deep neural network predicting the SPPB score was linking the IMU data to the ICF item d460: Moving around different locations and the deep neural network predicting the TUG score is linking to d450: Walking. Table 6.3 shows the association of the assessment and ICF scores. When one of the two deep neural networks detected a change, e.g. TUG score from 1 to 2 and so from ICF score 0 to 2, a 1 would be propagated to the metric function. When no change was detected a 0 would be propagated. The component also provided the information if the change was caused by an improvement or deterioration. The component is described in greater detail in reference [31].

Component 2: body functions. Component 2 measured the change in body weight by computing the time spent for preparing meals. The time was computed using the home automation and power consumption sensors attached to appliances

in the kitchen. The meal preparation time showed a strong significant Spearman correlation of at least |0.99| with body weight and HGS. Moreover, the model could detect weight changes of 5% in 3 months as an indicator for malnutrition in older adults [40]. It was not possible to draw inferences from the algebraic sign of the correlation coefficient about if the participant gained or lost body weight and the magnitude of the body weight change. Therefore, the component's output was 1 in case the body weight changed and 0 otherwise. Changes smaller than 1 kg were considered as normal fluctuations and were discarded during the component's built process. The result was linked to the ICF item b530: Weight maintenance functions. The component was not available for participants who were not preparing meals themselves at home. The component is addressed in depth in reference [32].

Table 6.3 The linking of the assessment scores to the ICF scores

Type/level of impairment	No	Mild	Moderate	Severe	Complete
SPPB score	12, 11, 10	9, 8, 7	6, 5, 4	3, 2, 1	0
TUG score	1	–	2, 3	–	4
ICF score	0	1	2	3	4

Component 3: environmental factors. The third component was comprised of an unsupervised concept drift detection machine learning model and a mathematical model. The unsupervised concept drift detection model detected normal and abnormal days. The features of the model were activity probability maps built from home automation and power consumption sensors. One map contained the probability a sensor events occurs in a certain hour of the day. The average probability map of 7 days was used as baseline. The machine learning model evaluated the probability map of a new day against the baseline. If the probability that the map of the new day was from the same probability distribution was smaller than a given threshold, the day was considered as normal and abnormal otherwise. The baseline was updated with each normal day to adapt to the behavior of the person. Figure 6.4 shows an example activity probability map [34]. Since the activity probability maps included without limitation the use of appliances and devices in the apartment and the output was linked to e115: Products and technology for personal use in daily life.

The mathematical model was a weighted directed graph representing the motion sensors in the apartment of the older adult. Each vertex of the graph

represented one room of the apartment and the edges where the possible transitions between the rooms are. The weights were the transition counts and used as activity measure. Every day was compared to the baseline and the difference in percent was computed. Two linear functions were fitted to the differences. The first linear function to the values of the control group and the second linear function to the values of the intervention group. The equations of the fitted lines are shown in Eqs. (6.2) and (6.3).

$$f_{control}(x) = -8.5829 \times 10^{-4}x + 0.1340 \tag{6.2}$$

$$f_{\text{intervention}}(x) = -4.9948 \times 10^{-5}x - 1.0610 \times 10^{-2} \tag{6.3}$$

where x is a period in days counted from the beginning of the study (T0). The results showed that the activity of the intervention group was decreasing slightly, whereas the activity of the control group was decreasing stronger by a factor of 10. The result of the sensor graph was linked to the ICF item e245: Time–related changes.

The outputs of both models were combined by the following equation

$$f_{env}(x) = |s| + \frac{c}{p} \tag{6.4}$$

where s is the slope factor of equations (6.2) and (6.3), c the abnormal day count, and p the time period T to T+1 the abnormal days were counted in. If f_{env} was greater than a given threshold, component 3's output was 1. For this research 0.3 was used. In clinical practice the threshold would be adjusted online based on the number of false positive alarms. The slope factor was chosen so, that the component was more sensitive towards a change for people not following any exercise programme. If no information were available s was set to 0.

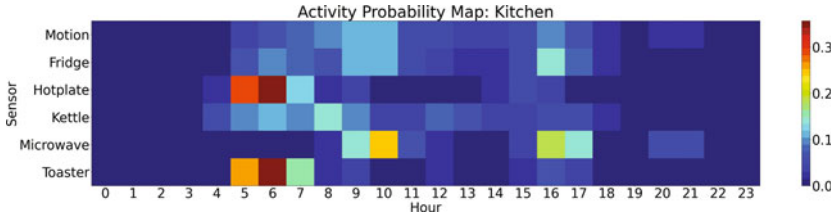


Figure 6.4 Activity probability map of participant 1's kitchen for day 1 [34]

The result could not be directly mapped to an ICF score and the output was indicating a change. A detailed description of the mathematical model for measuring e245 was published in [33] and the unsupervised concept drift detection machine learning model in [34].

Ethics Approval The methods were performed in accordance with relevant guidelines and regulations and approved by Institutional Review Board of the Carl von Ossietzky University. (protocol code: Drs.27/2014, date of approval: 30.04.2014).

Consent to participate. Written informed consent was obtained from all subjects involved in the study.

6.4 Results

This section is comprised of two parts. In the first part the general results are explained and in the second part the two study participants "Mr. Mueller" and "Mrs. Winter", whose names were changed due to privacy reasons, are examined in more detail and the application in physiotherapy is illustrated. The illustration is complemented by Figs. 6.5 and 6.6 showing the output of the system. The outputs of the remaining 18 participants can be found in the supplementary materials 6. The system raised an alarm every time a change occurred, even though not all components were available for each participant. Components 1 and 3 were available for each participant. Component 2 was not available for participants Mr. Mueller, 3, 4, 7, 9, 10, 12, 13, 14, and 19. These eight participants did not prepare meals themselves or had assistance. Table 6.4 shows the available ICF items, i.e. components for each participant. We used the assessment results to compensate missing IMU data in component 1 and still had missing values for participants 1, 7, 10, 11, and 15. The values were missing, because the assessment could not be performed due to their medical condition. Participant 14 was a special case, because the participant got cancer, underwent chemotherapy, and passed away. For component 3 there were only participant 7, and 14 with missing values. Participant 7 had a missing value in month 6. The participant was a special case as well, because he or she was hospitalised and moved to a nursing home after being discharged in month 6. Participant 14 had a missing value in month 2 due to medical condition. The cases of Mr. Mueller and Ms. Winter are described below as examples of how the system could be used by a licensed physical therapist to monitor the effect of an exercise programme to improve mobility and reduce the risk of falling.

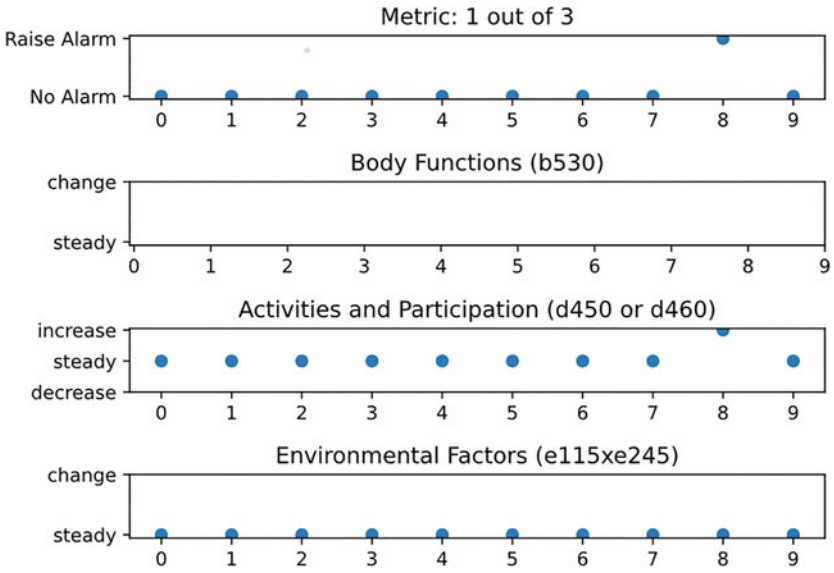


Figure 6.5 The system outputs of Mr. Mueller for month 0–9

Mr. Mueller was an 84 years old widower who sufferd from arterial hypertension, hypertensive heart disease, and subdural hemorrhage caused by a severe fall. He was living independently and was regularly supported by his children. The support included weekly grocery shopping, as well as transportation to medical appointments, but beyond that, he had no other contacts. Almost every day, one of the two children personally checked on their father. Mr. Mueller was pre-frail (3 pts. according to Fried's Frailty Scale) [36]. At the beginning of the study, he scored 3 in the SPPB mobility test as well as in the TUG with 31.63 s [1, 38]. Thus, he was considered at risk of falling and fall prevention training was advised. Following the OTAGO fall prevention exercise programme, Mr. Mueller received weekly home visits for the first two months and monthly visits thereafter. During these visits, he learned strength and balance exercises and received information about their importance for his everyday life. In addition to the exercises, Mr. Mueller was asked to take daily walks. Walking has never been Mr. Mueller's favourite activity. Therefore, he seldom took any in the last few years. In the beginning, he managed to walk with his walking stick for 5 min to the corner of the house. After 10 months he was able to walk 30 min with

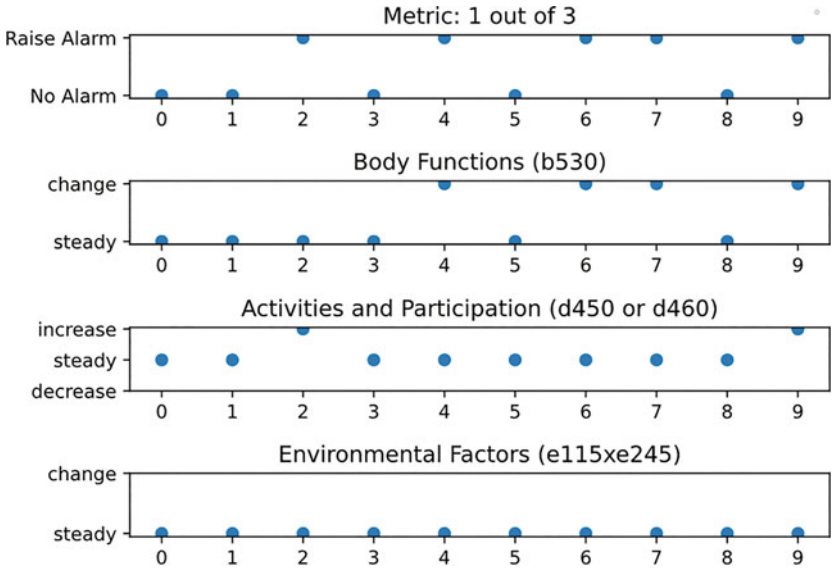


Figure 6.6 The system outputs of Mrs. Winter for month 0–9

his walking stick. As can be seen in Figure 6.1, there was no improvement or deterioration that caused an alarm up to month 8. The physiotherapist thought that this reflected Mr. Mueller's regular routines with weekly shopping, recurring household activities, visits by the children, and fall prevention exercises. Slowly increasing the exercise programme's intensity seemed to maintain functionality rather than significantly decreasing or increasing it. This was also evident in the mobility assessments, which were closely linked to the Activity and Participation component. In the SPPB, Mr. Mueller increased from 3 to 4 pts. over time and in the TUG he remained in category 3 but with an improvement from 31.63 to 22.31 s. In the third month, after cataract surgery, Mr. Mueller did not take any walks and fall prevention exercises for 2.5 weeks. However, this had no effect on the metric as can be seen in Fig. 6.5. The Body Functions component was deactivated for Mr. Mueller, because he had irregular lunches with his daughter. In months 8, the alarm indicated an increase in the ICF category Activities and Participation. The increase could also be seen in an improvement in TUG from 24.72 to 21.87 s. This improvement was relevant as it was above the Minimal Detectable Change (MDC) of 2.08 s [41]. In the SPPB, a re-improvement from 2 to 3 pts.

points could be seen. The physiotherapist evaluated the changes as positive and increased the exercise programme's intensity by increasing the walking time and adjusting the weights. Since the study ended no additional measures were taken. In a real-world case, the physiotherapist would tell Mr. Mueller to continue the exercises for the next 3 months without supervision and call if problems arise. After the 3rd month a followup examination would be done. However, since Mr. Mueller was not very motivated he might give less effort without supervision.

Table 6.4 The available components, i.e. ICF items, for each participant

ID	ICF items
Mr. Mueller	d450, d460, e115
Mrs. Winter	d450, d460, b530, e115, e245
1	d450, d460, b530, e115, e245
2	d450, d460, b530, e115, e245
3	d450, d460, e115, e245
4	d450, d460, e115, e245
5	d450, d460, b530, e115, e245
6	d450, d460, b530, e115, e245
7	d450, d460, e115
8	d450, d460, b530, e115, e245
9	d450, d460, e115, e245
10	d450, d460, e115, e245
11	d450, d460, b530, e115
12	d450, d460, e115, e245
13	d450, d460, e115, e245
14	d450, d460, e115
15	d450, d460, b530, e115, e245
16	d450, d460, b530, e115, e245
17	d450, d460, e115, e245
18	d450, d460, b530, e115, e245

Mrs. Winter (Fig. 6.6) was a 76 years old women with arterial hypertension and arthrosis in hips and knees who was living in assisted-living scheme. She was self-sufficient except for daily lunch at the facility but she prepared all other meals by herself. In the beginning of the fall prevention training, she scored 5 pts. in the SPPB and 3 pts. in TUG with 24.06 s. Over the 10 months, she

improved and ended the fall prevention programme with 10 pts. in the SPPB and in category 2 in the TUG with 11.90 s. In months 2, 4, 6, 7, and 9 the system alerted the physiotherapist. The body functions component indicated a change in months 4, 6, and 7. However, checking the weight with Mrs. Winter and asking about self-care revealed no abnormalities and was identified as normal fluctuations by a nutritionist. Alarmed by the metric, the physiotherapist noted an increase in the Activities and Participation component in month 2 and month 9. In month 2, the SPPB score changed from 4 to 6 pts. And in month 9 the SPPB score changed from 8 to 9 pts. The TUG remained in category 2 but increased from 15.40 to 13.12 s. The increase indicated that the right time had come for the physiotherapist to adjust the programme. The walking time could not be increased since Mrs. Winter was already taking long walks for 60 min. The intensity was raised by increasing the weight during strength exercises. After the 10 months, the physiotherapist ended the fall prevention training with Mrs. Winter. In a real-world case the physiotherapist would suggest that Mrs. Winter should continue the exercises on the current intensity level on her own and calls if problems arise. A follow-up in about 6 weeks would be done.

6.5 Discussion

OdAMS proved its usefulness for creating individualised rehabilitation programmes. The results showed that the component 1 for Activities and Participation was the most accurate one and delivered the most detailed information. Moreover, analysing the metric showed that none of the components was redundant. There were alarms triggered by a single component only. With privacy in mind, it would have been good to identify redundant components and remove them according to the data economy principle. In addition, it is important that all data is stored and processed on a computer in the flat of the older adult. Only the alarm should be sent to a server via the internet. Even though, the system's computing complexity was $\mathcal{O}(n^2)$ and dominated by the machine learning algorithms and the matrix multiplications inside them, OdAMS could be run on an ordinary computer. However, that might change in the future if the system is extended by new components. There are a few issues that must be addressed before the system can be used in clinical practice.

One issue is that the data was collected during pre-clinical pilot intervention study and the cohort size was small. Therefore, the components must be evaluated in separate clinical studies designed for each specific purpose. At this stage of the research, it is not possible to draw conclusions and the results can only

be considered indicative. Another point is that component 1 also needs some adjustments. The machine learning models processed intervals of 1 s length. A new metric, e.g. a majority vote ensemble, could be implemented to output a single score over a certain interval, because a change in mobility in a 1 s interval would be too sensitive and thus trigger too many alarms and is not meaningful. The shortest interval is 1 s and the longest has infinite length. However, infinite length is more of theoretical interest. The interval can be chosen according to the opinion of a physiotherapist, e.g. if high sensitivity is needed a short interval could be chosen. Also, the machine learning models must be fine-tuned to the individual. In contrast to component 1, the components 2 and 3 detected changes instead of absolute ICF item scores. Component 2 was linked to Weight maintenance functions and detected changes instead of an absolute score of the item. The component neither gives the information if the older adult gained or lost weight. Moreover, the component was applicable for only a few participants; consequently, the findings must be very carefully interpreted and are of preliminary nature. Component 3 was detecting changes of the item e115: Products and technology for personal use in daily life. The scoring interval was + 4 to 4 from complete facilitator to complete barrier, but concluding a score from a probability map is not possible. If the component raised an alarm and the comparison of the activity probability map of the anomaly and the reference showed that the probability of using, e.g. the kettle has dropped to 0 the reason was not obvious. One reason could be that the kettle was broken and the new one had not yet been delivered. Another functional related reason could be that the older adult could not use the kettle anymore because he or she could not push the buttons, read the caption, or could not see the pictographs on the buttons. In the first case the older adult would still have been a facilitator (score: + 4) of the kettle and in the second case it would have been a barrier (score: 4). When an alarm is raised, the person in charge must interview the older adult to investigate the reason for the change. Those considerations lead to another important point, the complex effects of changing conditions, e.g. a change in item d450: Walking likely affects other items like d530: Toileting. Our system has the potential to incorporate this complexity in two ways in future work. The first way is to use more refined components. The aim of component 2 was to measure changing body weight as an indicator for malnutrition. Adding important nutrition related information like consumption quantities might add valuable information that allows deeper in sights [42]. Component 1 processed IMU data to predict the SPPB and TUG score. The SPPB is comprised of three items and the TUG can be separated into the three items, sit-to-stand, walk, and balance. Since the component only predicted the overall score it was not possible to determine the contribution of

each item. A more refined version of the component could predict the subscores of each item to gain more insights. The second way is to use early fusion of the data and middle fusion of the components. An early fusion of the sensor data enables a more accurate estimation of parameters of interest, e.g. the fusion of IMU and motion sensor data allows a precise (i.e. smaller than the Minimal Clinically Important Difference (MCID)) estimation of the gait speed in domestic environments [43]. Since a lot of information is hidden in component 3 and its the activity probability maps a middle fusion with component 1 can give deeper insights in the impairment in daily life caused by reduced mobility. Fusing both components might reveal which part of daily life is affected the most, e.g. a decrease in mobility might lead to less activity on the upper floors of a flat and the resident may not be able to go to the balcony for recreation anymore.

Besides the pre-clinical study and component focused aspects, the usability and the effect on the daily practice of clinicians must be evaluated as well. If the sensitivity is too high many false alarms are raised and that would lead to dwindling acceptance amongst clinicians.

6.6 Conclusion

We introduced the system OdAMS for monitoring the functional status of older adults in daily life. The system linked IMU and home automation sensor data to three different ICF categories and raises an alarm if a change in one category was detected. The application to data of 20 (pre-)frail older adults showed the validity of the system. Moreover, we showed how OdAMS would be used by physiotherapists in two exemplary cases. Combining smart-home sensor systems with the ICF enables clinicians to tailor rehabilitation programmes to individuals and thereby improve their outcomes. Nevertheless, the system could be beneficial for the entire multi-disciplinary team—including physicians, nutritionists, and pharmacists amongst others. Our results are encouraging and future research should be directed to the possible impact on research and society.

OdAMS's potential impact on research is to enable researchers to conduct high-quality, reproducible, and transparent interdisciplinary research. The ICF provides a common language for describing and classifying a person's functional abilities across multiple disciplines. Using the ICF categories and items as a basis allows medical professionals from different fields to collaborate and identify relations between different medical aspects. For example, if the component 2 and component 3 are always raising alarms at the same time, that would be an indication for a relation between ICF items b530 weight maintenance functions

and e115/e245 products and technology for personal use/time related changes. Regarding interdisciplinary research between physician scientists and computer scientists, the linking of data to the ICF items bridges the gap between disciplines. Physician scientists could choose the ICF category and the specific items they want to investigate and what information they need. Based on that decision they can involve computer scientists to choose the data and the sensors according to their needs.

In society, the system has the potential to improve the quality of life of older adults by enabling them to living longer independently and to meet their psychological needs. In addition, the system enables physicians and medical professionals to apply individualised care and rehabilitation interventions. This would improve the well-being and physical condition of older people. It also supports the goal of rehabilitation to enable older people to maintain their independent daily lives. Therefore, the system supports physicians at assessing and keeping track of functional abilities and the effect of exercise programmes.

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Code availability The source code is available in the supplementary materials for academic purposes

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General Discussion

7

All the components and the system itself successfully passed the evaluation on the real-world dataset collected during the OTAGO study. The machine learning algorithms achieved high performance values and proofed to be precise using only a small amount of data. The traditional methods (i.e. linear regression) provided profound insights into the nutritional-, functional-, and mobility status. The example use case involving a licensed physiotherapist showed that the system was providing valuable information for creating individualised rehabilitation programmes and monitoring their progress; however, the introduced system was a proof of concept and further validation is necessary. Decision support is the main purpose of the system and it is not meant to be used as a standalone medical system, it would require a special certification acquiring is complex and tedious. Consequently, the system keeps a human in the loop and the human has to contact the older adult when the system raises an alarm. Following a common strategy in medical diagnosis, combining the system, which has a high sensitivity (i.e. high true positive rate), with a human, who has a high specificity (i.e. true negative rate), results in a high Receiver Operating Characteristics (ROC) value and hence in an accurate system. Besides the usability the design of the system must be changed before being deployed in a real-world application or study. The most important step is the validation of the components in clinical studies. The OTAGO study was designed as observational randomised pilot intervention study and the resulting cohort size of 20 was small. The results of the components one, two, and three were only indicators for a relation between the data and the ICF item.

The ICF category *Activites and Participation*, i.e. ICF items d460: *Walking* and d460: *Moving around different locations*, was measured by the component predicting SPPB and TUG scores using IMU data. While the deep neural network achieved an accuracy of 94.28% and 95.79%, the performance on a participant

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that was unknown to the network, was significantly lower. It is noteworthy that the component was predicting capacity on performance data. All labels were scores from mobility assessments supervised by medical professionals. Naturally, assessments are measuring the capacity because participants tend to give their best effort in testing situations. As a result, the test results rather reflect the capacity than the performance in daily life, e.g. participants are not always moving at their maximum pace in daily life. Also, the capacity is affected by the form of the day. During the preparation of the dataset, the assessment score, which was reflecting the capacity, was assigned to data, which was reflecting the performance, collected in daily life. That enabled the deep neural network to learn how to predict the capacity on performance data. Since a certain level of performance may be sufficient for daily life activities, participants with different capacity levels may have shown the same performance and in combination with the small cohort this may explain the low performance on unkown participants. The left out participants likely showed a combination of performance and capacity that was not present among the other participants. Despite the good results, they must be taken with a pinch of salt. The participants were responsible for charging the IMU, transferring the data, and carrying it with them. That not only led to faulty and incomplete data but also to an unknown position of the IMU on the participant's body. Moreover, there was a possibility that the participant did not carry the IMU at all. Another drawback was that the dataset did not contain all SPPB scores. As a result, the network could not detect if the score of a participant changed from a known to an unknown class, the network would predict one of its known scores. The model cannot give any information about the reasons for a score change. A medical professional must be involved as described in Chapter 6.

The afore mentioned challenges could be addressed in the following ways. Fine-tuning the network is one solution to improve the performance on unkown participants and it needs much less data than training it from scratch. Also, fine-tuning could be used to integrate scores, which were not present in the dataset, into the network. To detect if the IMU data shifts to an unknown score, the unsupervised concept drift detection approach in Chapter 5 could be used. When a drift is detected, the network could be fine-tuned on the new data and the new label could be assigned by medical professional afterwards. The unknown position could be mitigated by computing the magnitude of the data. Detecting if a participant was carrying the IMU or not may be more difficult because the motion characteristics of a sitting participant and an IMU lying on a table is likely, moreover, small motions may be superimposed by noise. Adding more information about the motion before and after could provide valuable information to distinguish between the two states. Regarding the handling of older adults,

the handling of the sensor must be simplified by reducing the complexity of charging the sensor and transferring the data. Inductive charging and automatic data transfer would lead to less missing data. Also, the sensor should be attached to something which older adults frequently carry with them—e.g. a certain piece of clothes. A clinical study would provide the controlled environment to have a detailled and gapless view on the participant and the sensor usage.

Component 2 was linking the data of the ambient sensors to the ICF item b530: *Weight maintenance functions* from the category *Body Functions*. As KPI the meal preparation time was estimated based on the sensor data from the sensors installed in the kitchen. Then, on a monthly interval, the trend of the estimated meal preparation time was correlated with the change in body weight. The significant Spearman correlation coefficient was $\leq |0.99|$ and all MCIDs were detected. The sign of the correlation coefficient did not allow to draw conclusions about gaining or losing weight. Furthermore, all changes in body weight of 5% or more in 3 months, which is considered as one indicator for malnutrition, were detected as well. Although all MCIDs and malnutrition indications were detected, the results must be considered as preliminary. Due to the heterogenous sensor setup of the participants, the estimated KPI differs in quality, e.g. some participants have had installed one motion sensor in the kitchen and one power consumption at the kettle, whereas other participants have had a second motion sensor over the stove and additional power consumption sensors at the microwave, coffee maker, etc. In the second case, as a result, the estimation was more accurate than in the first case. Another disadvantage was that it was difficult if not impossible to determine if the participants were preparing their own meals. As a matter of course, the component was not applicable for participants not preparing their meals on their own—receiving meals on wheels, having assistance while cooking—while the first information could be reliably retrieved from the diaries, there were no reliable information on the assistance by relatives. That may lead to estimations which were not reflecting the preparation time of the participant. Naturally, the component could not infer the intention of the weight loss; ESPEN guidelines are referring to unintended weight loss. One participant was weighting over 115kg and if the participant is put on a diet and looses weight, this would be an intended loss and no reason for any concern. It is necessary, as explained in Chapter 6 that a human is kept in the loop to determine the cause of the weight change. Nevertheless, the results were preliminary and only indication for the relationship between the meal preparation time and the body weight. To generate deeper insights and robust results a clinical study specifically designed for that purpose must be conducted. A clinical study design would ensure that all participants

have the same set of sensors installed in their flats and that the meal preparation time would be precisely recorded.

From the ICF category *Environmental Factors* two items—items e115: *Products and technology for personal use in daily life* and e245: *Time-related changes*—were measured by component 3, which consisted of two sub-components. Item e115 was measured by the sub-component described in Chapter 5 and item e245 by the sub-component described in Chapter 4.

For measuring *Products and technology for personal use in daily life* the motion and the power consumption sensor data were transformed to an activity probability map, which contained the probability of having a sensor event for the hours of the day. The maps contain the information about the utilisation of the appliances and the activities of the participant. Then, a variational autoencoder in combination with a statistical hypothesis test was used to detect abnormal days. The sub-component measuring e115 controlled the sensitivity of the first sub-component. Since people tend to forget things, only severe incidents and medical conditions were considered as reliable entries. Due to the unreliable and missing labels, evaluating the e115 sub-component was only possible to a limited extent. While all severe incidents, like severe medical condition, were accurately detected, i.e. classified as abnormal days, for many other detection no labels were available. There may were false positive detections amongst the days without a label. Nevertheless, the results on a special participant, who suffered from cancer and underwent chemotherapy, gave strong indication that the sub-component worked. Chemotherapy has a strong negative side effects on the body and mind, thus, none normal day was detected after the first chemotherapy treatment. Yet, the reason for an abnormal day is not necessarily negative. A participant with a lively and active private life has many abnormal days as well, but the reasons are of positive nature. Naturally, the sub-component could not infer the reason for an abnormal day. In this regard, a human must be kept in the loop to find the reason by talking to the participant. The most sophisticated way to validate the e245 sub-component would be a clinical study as well. The unreliable labels were caused by the retrospectively (about 4 weeks) filling of diaries. People, especially older adults, tend to forget normal or common activities; therefore, activities and special events must be noted down immediately. When the labels become available the unsupervised algorithm could be reliably validated. Sub-component measuring e245 was controlling the sensitivity by setting the alarm threshold depending on time and whether a participant was following the OTAGO exercise programme. Participating in the OTAGO exercise programme, decreases the risk of falling and improves the mobility of older adults. As a result, they are likely to be more active and henceforth the maps show a greater variety. As

a measure of mobility, motion sensor events and transitions between rooms in the flat were used and the trend over time was analysed. Based on the ground plan the sensors were modelled as a graph, where the sensors were the vertrices and the transition routes the edges. The results showed that the activity of the exercise group slightly declined, whereas the activity of the other group declined stronger. Dependent on the time the threshold for raising an alarm was adapted. Consequently, the component was more sensitive for participants not following the exercise programme than for the participants in the exercise group. Not all false positive alarms could be averted; however, in the medical context a false positive is preferable over a false negative, which may result in missing a severe medical condition. Validating the sub-component was also difficult because of the way mobility was measured. Visitors and other people in the flat cause an overestimation of the activity and hence bias the result; thus, no causalities could be concluded. Conducting a sophisticated validation, e.g. in a clinical study, is needed to reveal any causalities. Close attention must be paid to reliable and accurate logging of the activities in the domestic environments and particularly, the presence of other people. In conjunction the full potential of the sensor graph could be exploited. Medical conditions affecting the behaviour may be observed in the graph and number of transitions, e.g. frequent bathroom visits may indicate a medical condition related to the bladder and the digestive system.

Last but not least, the complete prototype should be evaluated in a study, focused on application and acceptance. When developing the prototype ethical and privacy as well as safety and security aspects must be considered. Data collected in the most private spaces and the personal living space of people is worthy of protection. Additionally, the legislature acknowledges privacy as a fundamental right and implemented strict regulations by the means of the General Data Protection Regulation (GDPR) [8]. Five main ethical and privacy challenges for sensor systems installed in domestic environments have been identified [16, 2]. When dealing with systems collecting information and data, the first which comes to mind is the informational privacy, in other terms the right to control which information are recorded, the information flow, and the information access. The proposed system to some extent respects information privacy by design. Which information are recorded can be controlled by choosing the modules and sensors one feels comfortable with having them installed in the private space. Also, the flow of information can be minimised by using an on-premise rather than a cloud system. All data or information would be stored on a local machine in the home and this makes data and information transfer to any third party unnecessary, which reduces the risk of data breaches. Information access can be controlled by choosing the recipient. Nonetheless, at a certain level the system

becomes unusable, e.g. if one does not want any sensor installed in their home, the system will not work. Secondly, physical privacy must be taken into consideration. Physical privacy includes but is not limited to solitude, bodily modesty, and bodily integrity. On the one hand, using ambient sensors and deliberately abandon cameras, protects physical privacy because people do not have to leave their homes for certain medical checks. On the other hand, the impending contact by an alarmed relative or medical professional undermines solitude and people may feel like somebody is always around monitoring them in the form of the sensor system. While alarming only the resident and nobody else protects solitude, the usefulness is questionable because the decision whether to take actions on an alarm would be left to the resident and not to a medical professional. Thirdly, including and excluding certain people from intimate personal experiences, like suffering and recovery, is defined as associative privacy. Usually, people want to include medical professionals in their process of recovery or suffering and the system was designed to exactly do that; however, the level of inclusion can be controlled by choosing which components are installed and which alarms are raised and sent. As before this can be done to a certain level before the system becomes unusable. Not allowing any alarms will make the system meaningless. The fourth privacy dimension, proprietary privacy, is closely related to physical privacy. People are the sole holder of the ownership of their bodies and its products. Certain sensors could be used to retrieve biomarkers, such as DNA and fingerprints. Consequently, those sensors and information could be used for monetarisation purposes and identity theft. At the current development stage, the system is not detrimental to proprietary privacy because it is solely using ambient and wearable sensors; however, further development of new components may add new sensors affecting proprietary privacy. Fifth and last dimension is the decisional privacy, which refers to the active participation in making decisions without the interference of other people and authorities. Again, this is an important but difficult aspect because the system was designed to enable medical professionals to influence the decisions of their patients. Having the best of intentions, medical professionals will have a negative impact on the decisional privacy of the patient. While the influence may be in the best interest and beneficial for the health of the patient, the right of decisional privacy must not be neglected. Diminishing the decisional privacy must be reserved for exceptional circumstances and with due regard to the patients will and needs.

Considering the application area, the system must be safe and secure by design. The term safety refers to a safe—it causes no harm—use of the system and its software and the term security refers to the prevention of unwanted or illegal activity in the system and its software. Sensor related safety can be achieved

by properly installing them in the flats so that motion sensors cannot fall off and the power consumption sensors cannot cause any electrical shocks. Another important point is that the sensors are not interfering with the daily life of the resident because interfering sensors may be moved by the resident and moving could cause harm to the resident. Furthermore, sensors should not be obstacles, which can lead to falling, and this also holds for the base station. The resident can even stumble upon a small computer and hence it must be stored in a special place like a cabinet. In the current development stage, the system has no human interaction component for the resident, but it is preferable to implement such a feature in the next version. Through an interface the resident could control the software to a certain extent and be in charge of his own privacy rights. Since the software should be designed for the target group of older adults, special attention must be paid to their specific needs, e.g. large text and icons are preferrable over command line interfaces. Yet, there is no gold standard for designing interfaces for older adults and the best way for designing is iteratively keeping the user in the loop [12]. Besides the usability, the user should also not be able to cause crashes, intentionally or unintentionally. Moreover, the correctness of the software must be ensured, this can be done by following software development guidelines, coding standards and formal verification. Currently, the modules are all implemented in Python following the PEP8 code style guidelines and the imperative programming paradigm. That makes the implementation understandable, easy to extent, and easy to maintain for other developers. Moreover, the software is operating system agnostic and can be used with either Microsoft Windows or Linux, which is open source and offers distributions designed for being resilient and secure. Moreover, this also pays off on security as well. Another common test strategy is having software testers who are not the developers. They can do comprehensive tests and this miminises the risk of software bugs, which could be used for malicious actions by attackers. Running the whole system on—premise and minimising data sending reduces the risk of data breaches; however, the system would not be scalable anymore. If new components, which exceed the currently available computational resources are added, the hardware must be changed by a technician. In a cloud computing scheme using infrastructure as code more hardware resources can be added with small effort. Even though the communication over the internet is minimised, it must be secured. Virtual Private Networks (VPN) are one way to provide a secure connection for sending the alarms over the internet. All other communications over the internet must be blocked. Consequently, the security level is high, but on the expense on maintainability and upgradability. When updates are adviced or new components are supposed to be added, a technician has to do it on site. Not only the software must be secure, but

also the sensors. Sensors are collecting data and transfer it to the base station. If the data is transfered over a wireless connection it must be secured by encryption. Another secure option, which is more difficult to install, is to use wired sensors. The next step would be to secure the base station. To prevent unauthorised people from access, it should stored in a secure place, e.g. a locked cabinet. Also, the user must be authenticated before accessing the software and retrieve information. Biometric authentication using face recognition or fingerprint readers are reliable and secure. A well known fact in cybersecurity is that the user is one of the highest security risks. As a fundamental rule, the resident should only give trustworthy people access to the system. For sake of simplicity, the number of people should be small and limited to relatives, technicians, and medical professionals. To make it even easier for older adults, one guardian, who decides about the access, can be named.

In medicine and health care cost efficiency is of great concern; consequently, the hardware of the system must be comprised of standardised cheap components. The sensors are ordinary motion and power consumption sensors and hence available at a low price, whereas the used IMU is too expensive for a real—world application and must be replaced by a cheaper version with similar characteristics. As base station an ordinary personal computer is sufficient. The most computationally expensive operations ($O(n^2)$) are the deep learning models of components 1 and 3. Usually, graphic cards or special hardware for machine learning inference provide the best performance for deep learning applications; however, both are expensive and not necessarily needed in this context. Real time computation is none of the requirements of the system and the latest central processing units (CPU) provide advanced instruction sets for vector and matrix computations, e.g. Advanced Vector Extensions (AVX) on Intel CPUs. Optimising the used deep learning library, e.g. by compiling TensorFlow with AVX support, to leverage the advanced speed of the special CPU instruction set significantly reduces the computation time compared to an unoptimised version. While optimising the libraries is almost costless, buying additional hardware in the form of graphic cards is very costly.

The system has the potential of having an impact on science and society. Making use of the ICF as common language not only supports the collaboration between researchers of different medical fields but also enables high quality reproducable interdisciplinary research between medical scientists and scientists from other research disciplines, e.g. computer scientists. Medical researchers are now able to reveal unknown relations and interactions between different aspects by investigating the outputs of the components. If two components are always raising alarms at the same time, that would be an indicator for a relation between

the components and consequently between the ICF items and further analysis can be done based on those finding. Since the system is extendable by design, it is possible to add new components as needed. If a possible new relation should be investigated, the new component can be designed according to the research question and integrated in to the system. For the prototype, the data of the GPS receiver was not used. A possible research question would be if the meal preparation time is related to a person going to the farmers market. Intuitively, being able to go to the farmers market, the older adult may be able to prepare meals at home and preparing the fresh and unprocessed ingredients for cooking takes more time than preparing convenience meals. The new component would raise an alarm when the older adult was at the farmers market. The possible impact on society is to lessen the demand on the health care system and supporting older adults living an independent life and also increases their quality of life. Application in medicine and for individualised rehabilitation programmes in particular may lead to a faster, substantial, in a sustained manner recovery of older adults. As a result, lesser follow-up treatments and lesser care is needed. Moreover, monitoring the daily life of older adults, the system can give indication when an older adult needs support at day-to-day tasks and appropriate assistance could be given. As a consequence thereof, their well-being is improved by fullfilling their innate psychological needs and helps aging in place. During the COVID-19 pandemic social distancing was the order of the day and for older adults, as a corona high risk group, in particular. In some nursing homes the situation was difficult regarding visiting hours, infection events, and the impact on medical care. Especially in this situations, the proposed system could have been useful, even saving lives, by reducing the social interaction to a minimum and keeping older adults safe in their own homes. Since the system can be used to precisely identify where assistance is needed, simple assistance could be provided by relatives instead of professional caretakers; as a consequence, the risk of infection is minimised because a professional caretaker is likely to visit more people than the relatives. Also, isolation in the own homes is more effective than in a place where many people are living. Therefore, the system has not only the potential to increase the quality of life of older adults, but also to protect their lives.

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